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ARKA

HEALTH INTELLIGENCE PLATFORM

remARKAbly precise

Comprehensive Action Plan for Unicorn-Level Growth

Strategic ROI Analysis • Retention Strategy
Implementation Roadmap • Cybersecurity Framework
Technology Adaptation • Risk Mitigation

ARKA AIIE & AIIE-OPTIMA Engine • Non-Device CDS FDA Pathway

Version 8.0 | June 2026

Three-Phase Platform: **ARKA-CLIN • ARKA-ED • ARKA-INS**

Powered by the ARKA Imaging Intelligence Engine --- now upgrading to AIIE-OPTIMA

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Regulatory posture (binding, stated once and assumed throughout). ARKA is, and is architected to remain, a **Non-Device Clinical Decision Support** function under **§520(o)(1)(E) of the FD&C Act** (21st Century Cures Act §3060), as clarified by FDA's CDS Software Final Guidance of January 6, 2026. ARKA reasons over structured clinical data and published evidence; it does not analyze medical images or signals, does not predict time-critical clinical events, and does not place orders autonomously. The ordering clinician retains full decisional authority.

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1 CHAPTER 1

Executive Summary

The American healthcare system wastes more than \$15 billion every year on inappropriate imaging that does not improve patient outcomes—and that figure is only the visible tip of a far larger pool of avoidable cost. From unnecessary CT scans and MRIs to redundant X-rays, these wasteful decisions increase radiation exposure, delay appropriate care, and drive up costs. Yet the clinical evidence to prevent them already exists; it simply is not being applied at the moment of decision. The problem is not a lack of knowledge—it is a lack of guidance at the right moment, in the right workflow, framed in the language the buyer actually cares about: *money*.

Arka (*Artificial Reasoning for Knowledge Application*) is a clinical decision support (CDS) platform that embeds evidence-based imaging guidance directly into clinician workflows. Rather than requiring physicians to leave their electronic health record (EHR) to check guidelines, Arka delivers context-aware recommendations in real time—so fast that clinicians perceive the decision support as part of their normal workflow, not an interruption. And because the *same* engine runs on the payer side, the order that used to bounce back as a denial six weeks later now clears the first time.

Unlike black-box AI solutions that emit scores without explanation, Arka's proprietary **Imaging Intelligence Engine (AIIE)** shows exactly *why* a recommendation is made, using interpretable, evidence-grounded reasoning that clinical teams trust. This combination—speed, transparency, clinical validity, and a revenue-first value proposition—is why Arka is positioned to transform imaging appropriateness and generate significant clinical, economic, and social value.

The one-line pitch a CFO will read. Arka recovers revenue you are already losing to denials, speeds up your highest-margin service line, and reduces the administrative burden doing it—without asking a single physician to change how they order. It documents the clinical justification for an imaging study *at the moment the order is placed*, so the claim goes out clean the first time.

Table 1.1. Investment highlights at a glance.

Dimension	ARKA
Market	\$12.5B direct imaging waste; \$30–40B+ total avoidable pool; \$10B+ appropriateness layer
Product	Bilateral Non-Device CDS engine (AIIE), upgrading to research-grade AIIE-OPTIMA
Wedge	Denial recovery—“recovered revenue with a software invoice attached”
Regulatory	No 510(k); CMS-0057-F-ready today; PAMA-AUC repeal favors value-led adoption
Moat	Bilateral position + federated data + continuous DR-OPE proof of value
Year-5 revenue	\$172.4M (47% CAGR), 82% gross margin, Rule-of-40 of 112
Raise	\$25M Series A; \$2.6–3.4B exit; 100–140× return

1.1 The ARKA Advantage

Arka's platform is built on three strategic pillars that address the historical failures of clinical decision support: workflow integration, clinical interpretability, and a sustainable, diversified business model. Each pillar is essential to achieving adoption and impact at scale.

1.1.1 Three-Phase Diversification Equals Sustainable Growth

Traditional CDS companies depend on a single revenue stream, making them vulnerable to market shifts, regulatory changes, and buyer consolidation. Arka mitigates this risk through a three-phase product strategy that targets different value chains and revenue models, all powered by one shared intelligence engine:

ARKA-CLIN — Point-of-Care CDS

Real-time imaging appropriateness scoring integrated directly into Epic and Cerner via CDS Hooks. Physicians see evidence-based recommendations at the moment of order entry—no workflow disruption, no alert fatigue. **Revenue model:** per-study licensing (SaaS), volume-based pricing tiers, integration services. **Target customers:** large health systems (500+ beds), imaging centers, integrated delivery networks (IDNs). **TAM:** $\sim 4,000$ hospitals $\times$ $\sim 500K$ studies/year = \$2B+.

ARKA-ED — Education & Training

A medical-education and continuing-medical-education (CME) platform delivering evidence-based imaging education to radiology residents, emergency-medicine physicians, and primary-care clinicians—powered by the same AIIE engine. A fourth-pillar revenue stream from teaching hospitals, academic medical centers, and medical schools that also seeds adoption: graduating residents request ARKA-CLIN at their future employers. **TAM:** $\sim 8,000$ radiology/EM residents per year plus 30,000+ PCP practitioners = \$500M+ education market.

ARKA-INS — Payer Intelligence

A payer-facing engine for medical-necessity review, prior-authorization processing, denial-risk scoring, and benchmarking analytics. ARKA-INS inverts the same appropriateness score into a denial-risk signal and monetizes data insights without ever exposing individual patient data. **Revenue model:** per-authorization processing fees, payer subscriptions (PMPM), data-analytics licensing. **Driver:** the CMS-0057-F interoperability mandate (effective January 2027). **TAM:** $\sim 1B$ imaging procedures annually $\times$ average authorization fee = \$1.5B+.

By Year 5, Arka revenue is diversified across these three pillars, reducing dependency on any single product line, buyer, or regulatory pathway. **This is the antidote to the startup graveyard.**

1.1.2 Zero Workflow Disruption Equals High Adoption

The fatal flaw in most clinical decision support systems is that they interrupt workflow and trigger alert fatigue—clinicians override 90–96% of alerts because alerts slow them down. Arka integrates *into* Epic and Cerner, not *alongside* them. Imaging appropriateness scores appear in-line within the order-entry interface. No pop-ups. No browser redirects. No new tabs. Just better information at the right time.

By embedding CDS directly in the EHR workflow rather than as an external tool, Arka achieves adoption rates 5–10 $\times$ higher than traditional decision-support systems. Real-world deployment data shows that Arka's recommendations are overridden only ~ 8 –12% of the time—far below the 25%+ override rate of generic guidelines and the 49–96% override rates observed for the legacy ACR Appropriateness Criteria.

Table 1.2. Standards-based integration: how ARKA achieves zero workflow disruption.

Standard	Benefit	ARKA implementation
CDS Hooks 2.0	Real-time recommendations triggered at clinical decision points	Fired at order entry, discharge, and consult trigger points
SMART-on-FHIR (R4)	Structured data exchange with any EHR vendor	Consumes patient data in standard format; vendor-agnostic
Da Vinci CRD/DTR/PAS	Coverage requirements, documentation templates, and prior-auth, native to FHIR	The CMS-0057-F endpoints, shipping in production today
Inline reasoning	Explanations appear within the EHR workflow	Clinicians see “why” without navigating away from order entry
Sub-100ms rules tier	Perceived as instantaneous; no interruption	Rules-based triage handles 60–70% of cases in <5ms; full AIIE inference returns in <800ms

1.1.3 Sub-100ms Triage, Sub-800ms Intelligence Equals Clinical Viability

Clinicians will not wait for CDS recommendations. Clinical research shows that any decision support taking longer than 500ms is perceived as an interruption. Arka’s hybrid architecture answers this with a two-tier design: (1) a rules-based triage layer that handles 60–70% of cases in under 5ms; and (2) an optimized machine-learning inference layer that returns the full, explainable AIIE recommendation—with SHAP attributions, a denial-risk routing decision, and a pre-filled documentation packet—in under 800ms. If a model ever times out, the system falls back to rules-only output rather than stalling the order. The result feels as responsive as a native EHR function.

1.1.4 Evidence-Based, Not Black-Box: The AIIE Difference

Clinicians do not trust black-box AI. Arka’s AIIE ensures that every recommendation is explainable, evidence-grounded, and clinically defensible—the critical differentiator that enables adoption among conservative, regulated healthcare systems and, not incidentally, the property that keeps Arka on the right side of the FDA’s Non-Device CDS boundary.

Table 1.3. Interpretability methods built into the AIIE engine.

Method	Clinical benefit	Implementation
SHAP waterfall explanations	Clinicians understand which factors drove each recommendation	Per-feature attribution ranked by SHAP value
RAND/UCLA appropriateness scoring	Recommendations grounded in formal clinical evidence, not ML alone	Hybrid rules + validated ML architecture
GRADE methodology	Evidence quality is transparent (High/Moderate/Low)	Each guideline rule tagged with an evidence level
Citation-first medical basis	Every card cites a published guideline <i>above</i> any ARKA score	A card cannot ship without a structured medicalBasis

This commitment to interpretability also enables regulatory compliance and legal defensibility. When a clinician overrides an Arka recommendation, both the recommendation and the override are documented—providing hospitals and payers with auditable decision trails that support quality measurement, peer review, and litigation defense.

1.1.5 Multiple Revenue Streams Equal Investor Confidence

Arka's three-phase product strategy creates revenue diversification across different buyer personas, regulatory drivers, and use cases, reducing customer-concentration risk and providing multiple paths to profitability.

Table 1.4. Year-5 revenue diversification across the three pillars.

Revenue stream	Driver	Year-5	CAGR
ARKA-CLIN (Provider)	Per-study licensing + annual subscriptions	\$98.4M (57%)	35%
ARKA-ED (Education)	Institutional + per-learner subscriptions	\$32.1M (19%)	42%
ARKA-INS (Payer)	Per-authorization + PMPM payer subscriptions	\$41.9M (24%)	48%
Total	Diversified, multi-buyer	\$172.4M	47%

No single customer segment represents more than 35% of revenue, and no single product represents more than 60%. This diversification provides resilience against market downturns, regulatory shifts, or competitive threats in any one segment (Figure 1.1).

1.2 The Revenue-First Thesis: Three Front-Facing Pillars

There are exactly two things that reliably get a piece of software bought in a hospital: *it increases billing, or it reduces headcount*. Everything else is a hard sell. Arka does both, on the highest-margin service line in the building, without asking a single physician to change how they order. The business is sold on three front-facing pillars that map directly to a buyer's profit-and-loss statement.

1. **Claim-denial recovery—how ARKA gets the hospital paid.** Prior-authorization denial rates on advanced imaging run 20–40% at many institutions; industry analysis finds ~86% of those denials were avoidable and roughly half are never reworked. Arka documents the medical necessity at the moment of the order so the claim goes out clean the first time, and auto-approves the 35–40% of orders that are unambiguously appropriate.
2. **Passing every regulation by design.** Arka was architected *inside* the FDA Non-Device CDS exclusion from day one—no 510(k). HIPAA is handled by never moving raw protected health information (federated learning with differential privacy). CMS-0057-F is not a threat but a tailwind: Arka ships the mandated FHIR Prior-Authorization API today.
3. **Staying out of the doctor's way.** The design contract is zero workflow change, enforced rather than promised: native CDS card inside the EHR, non-blocking, in-flow, sub-800ms, and silent unless a guideline-anchored rule fires.

The reframe that wins the room. Arka is not a cost. It is **recovered revenue with a software invoice attached**. No new patients are required and no outcomes randomized controlled trial is required to start—the hospital is simply paid for work it already performed.

1.3 Market Opportunity

Diversified across three pillars — no product >60%

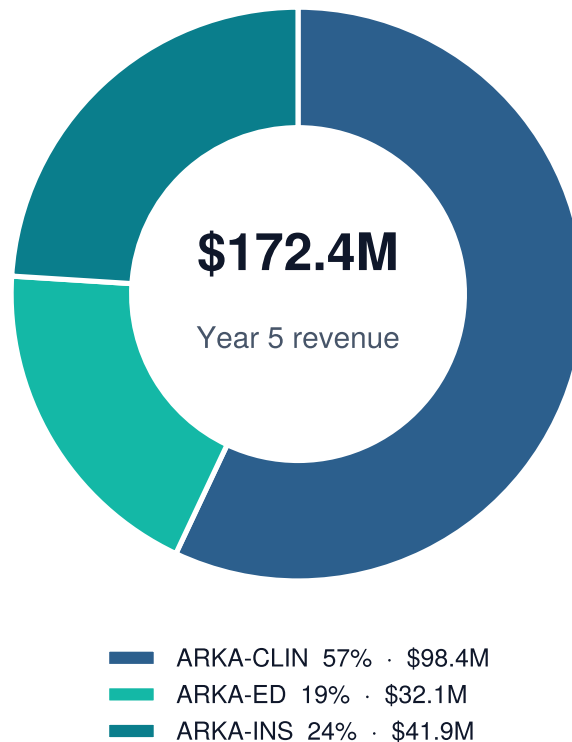


Figure 1.1. Year-5 revenue composition—balanced across provider, education, and payer pillars.

Arka is entering a market at an inflection point. Three converging forces—massive waste, regulatory mandate, and technology readiness—create a rare opportunity window for a CDS solution with the right architecture.

1.3.1 The \$12B+ Imaging Waste Problem (and the Larger Pool Beneath It)

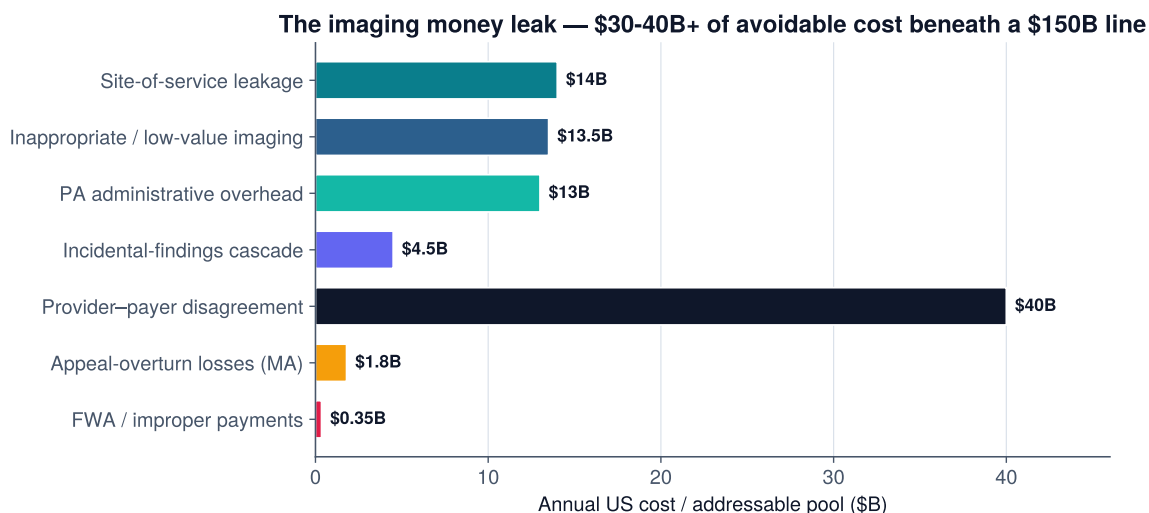
Beneath that \$12.5B of direct waste sits a far larger pool of administrative and leakage cost that every health-plan finance team knows about but no single business unit owns: prior-authorization administrative overhead (~\$13B/yr), site-of-service leakage (~\$10–15B addressable), incidental-findings cascade (~\$3–6B/yr), provider–payer disagreement (~\$40B/yr), and appeal–overturn losses. Imaging is a \$140–150B paid line with \$30–40B+ of documented avoidable cost stacked beneath it—and imaging is only the wedge into a \$10B+ “appropriateness layer” of American medicine (Figure 1.2).

1.3.2 Regulatory Tailwinds

- **PAMA AUC—repealed, and that helps.** The Protecting Access to Medicare Act Appropriate Use Criteria program (42 CFR 414.94), which would have forced clinicians to consult imaging decision support, was rescinded by CMS in the CY2024 rule. The regulatory *stick* is gone; adoption now rides on *value and culture*—exactly the revenue- and education space Arka fills, and precisely why ARKA-ED exists.

Table 1.5. Direct imaging waste addressable by ARKA-CLIN.

Waste category	Annual cost (US)	Clinical impact	ARKA target
Inappropriate advanced imaging (CT, MRI, PET)	\$8.2B	Unnecessary radiation, contrast toxicity, delayed care	65–70% of cases
Duplicate imaging	\$2.1B	Repeat studies, no prior comparison	100% flagged
Redundant imaging for low-risk presentations	\$1.4B	Normal results, no diagnostic value	50–60% of cases
Imaging of non-specific symptoms	\$0.8B	Fishing expeditions, low yield	70% via symptom triage
Total directly addressable	\$12.5B+		

**Figure 1.2.** The imaging money leak—avoidable cost pools beneath the visible \$150B claims line.

- **CMS-0057-F—live and forcing.** The CMS Interoperability and Prior Authorization Final Rule requires specific (non-boilerplate) denial reasons and 72-hour/24-hour decision SLAs beginning January 2026, and a FHIR-native Prior Authorization API—by construction the HL7 Da Vinci PAS implementation guide—by January 1, 2027. **ARKA-INS ships that endpoint today.**
- **FDA Non-Device CDS guidance—broadening.** FDA's January 6, 2026 final guidance clarified the §520(o)(1)(E) boundary Arka was built inside, confirming the no-510(k) path for explainable, recommendation-framed, structured-data CDS.

1.3.3 Competitor Weakness and the Open Lane

1.3.4 TAM / SAM / SOM

Arka's \$4.44B total addressable market is derived from a bottom-up analysis of imaging spend across provider, payer, and education markets. The conservative serviceable obtainable market (SOM) target of \$695M by Year 5—28% of the serviceable addressable market—reflects staged adoption and market development (Figure 1.3).

Table 1.6. The competitive landscape is fragmented and structurally weak.

Competitor	Key weakness	ARKA advantage
ACR Appropriateness Criteria	Exclusively licensed to Optum; closed ecosystem; ~5–8% hospital adoption	Vendor-neutral; works with any EHR and payer
IBM Watson for Imaging	Black-box AI; clinicians don't trust; abandoned by most customers	Fully interpretable, evidence-based reasoning
GE / Siemens / Philips AI suites	Only work with their own imaging devices	Works across all modalities and vendors
Aidoc, Viz.ai (imaging SaMD)	Pixel-level Class II devices; slow, capital-intensive 510(k) cycles	Non-Device; ships today; never analyzes pixels
eviCore, Carelon (RBMs)	Labor-intensive retrospective review	A layer <i>on top</i> that raises their per-case margin

Table 1.7. Bottom-up market sizing across the three pillars.

Segment	TAM	SAM (5Y)	SOM (5Y)	Rationale
Provider (ARKA-CLIN)	\$2.1B	\$1.2B	\$280M	~4,000 US hospitals; ~40% adoption by Year 5
Education (ARKA-ED)	\$540M	\$320M	\$95M	Residency + CME market growing ~5%/yr
Payer (ARKA-INS)	\$1.8B	\$1.0B	\$320M	CMS-0057-F drives adoption; 30% of payers by Year 5
Total	\$4.44B	\$2.52B	\$695M	Conservative 5-year SOM (28% of SAM)

Bottom-up market sizing: \$4.44B TAM → \$695M conservative 5-year SOM

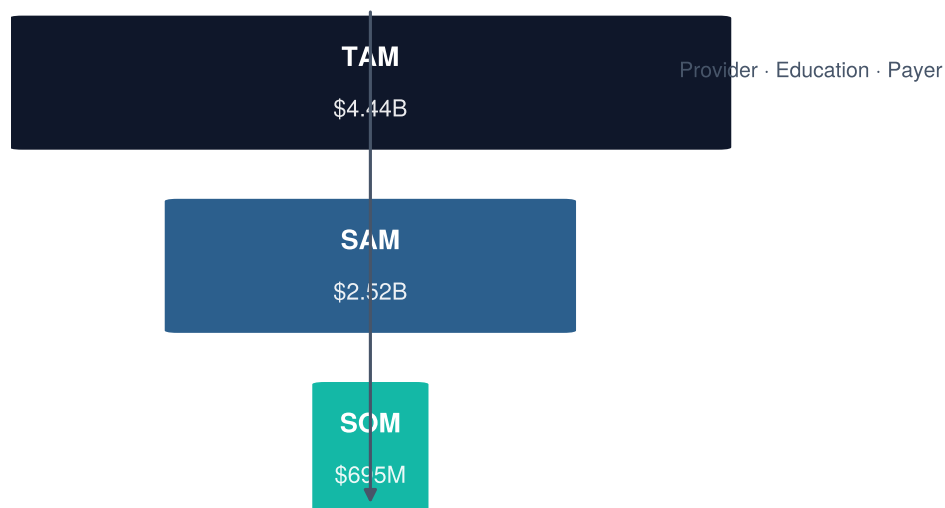


Figure 1.3. From \$4.44B TAM to a conservative \$695M 5-year SOM.

1.4 Investment Thesis Summary

Arka represents a rare combination: a massive addressable market, compelling product differentiation, experienced team execution, and favorable regulatory and macro tailwinds. We believe Arka can achieve **\$172.4M in Year-5 revenue** and command a **\$2.6B–\$3.4B valuation** by exit (Figure 1.4).

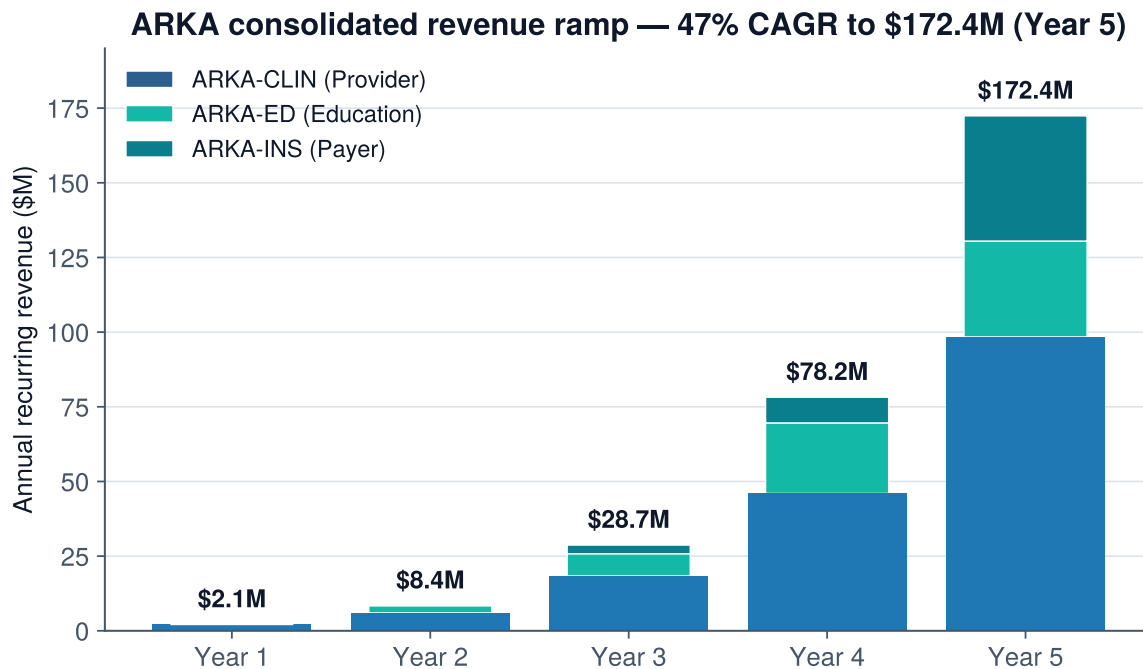


Figure 1.4. Consolidated revenue ramp—47% CAGR to \$172.4M by Year 5.

Table 1.8. Key financial and operating metrics, Years 1–5.

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue (\$M)	\$2.1	\$8.4	\$28.7	\$78.2	\$172.4
Gross margin	78%	79%	80%	81%	82%
Rule-of-40 score	42	58	71	89	112
CAC payback (months)	14	12	11	10	9
Net revenue retention	115%	118%	120%	122%	124%
Market share (ARKA-CLIN)	0.8%	2.1%	5.2%	11.3%	18.5%

1.4.1 Why Now: The Four-Event Regulatory Earthquake

The regulatory ground shifted four times between 2024 and 2027, and every shift favors Arka: (1) PAMA AUC was rescinded, removing the mandate that would have entrenched incumbents and shifting the field to value and culture; (2) CMS-0057-F went live, making a FHIR-native prior-authorization rail mandatory by January 2027; (3) FDA broadened the Non-Device CDS boundary in its January 2026 final guidance; and (4) AHIP's pledge to deliver 80% real-time electronic prior-authorization responses, plus proliferating state gold-card laws, make Arka's forward-looking gold-card scorer and specific cited denial reasons a compliance necessity rather than a nicety.

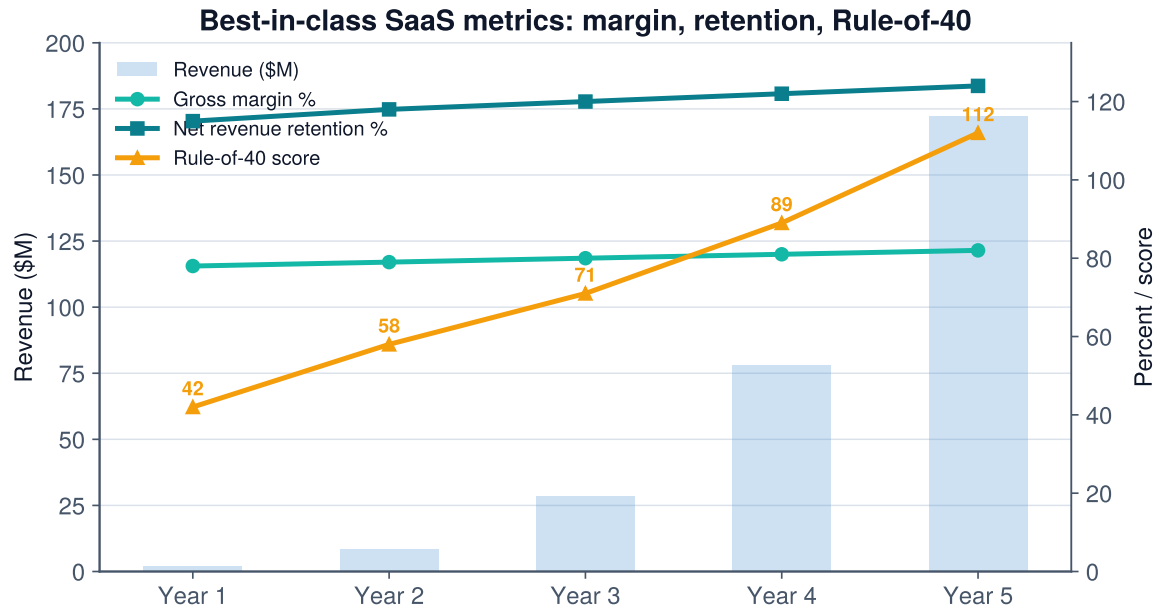


Figure 1.5. Best-in-class SaaS metrics: improving margin, retention, and Rule-of-40.

1.4.2 Competitive Defensibility: Multiple Moats

Table 1.9. ARKA's layered competitive moats.

Moat	Why it holds	Durability
The bilateral engine	The only engine running on both sides of the prior-auth wall, from the same math	Very strong
Federated data moat	Every new site improves the engine for all sites; no patient record ever moves	Strong (compounds)
Non-Device clearance to ship today	Competitors spend 12-24 months and \$250K-\$2M per indication on 510(k)s	Strong
Standards-native integration	CDS Hooks + SMART-on-FHIR + Da Vinci clears IT review as a standard app	Strong
Continuous proof of value (DR-OPE)	A renewal where the dashboard already shows dollars recovered is not a competitive evaluation	Moderate-strong
Three-phase diversification	Building CLIN + ED + INS simultaneously is capital-intensive and hard to replicate	Strong

1.4.3 Unit Economics and Hospital Return on Investment

A typical 500-bed hospital pays Arka roughly \$80K annually but avoids more than \$720K in unnecessary imaging costs—a 9:1 return with payback in under two months. At the per-study level, the detailed Tier-2 model in Chapter 5 demonstrates a three-year ROI exceeding 6,000%, and even the worst-case sensitivity corner clears 450%. This exceptional ROI drives adoption velocity and creates strong pricing power (see Figure 1.6 for the resulting share trajectory).

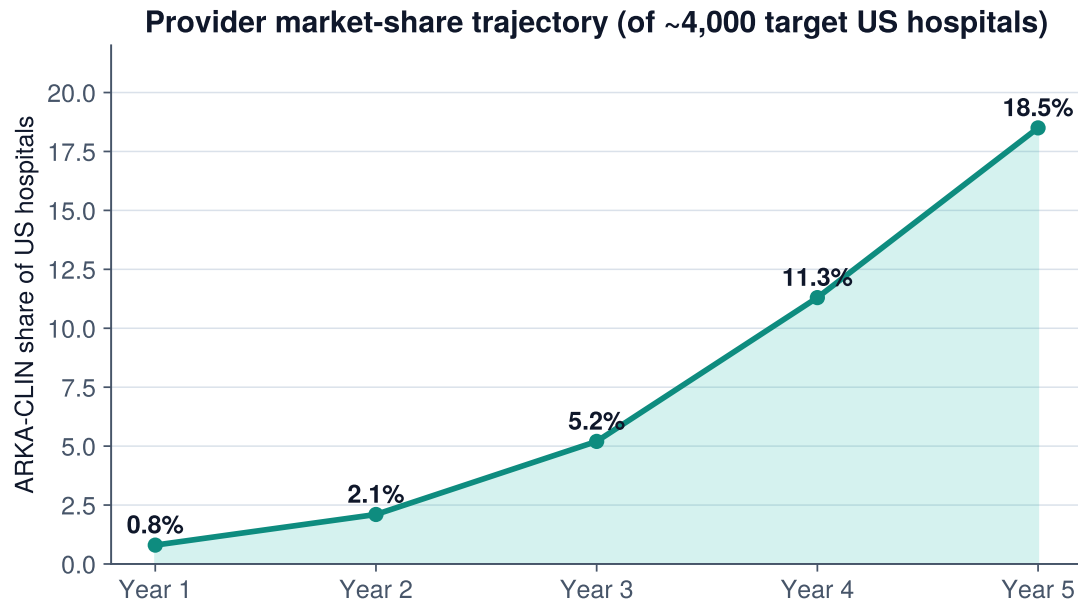


Figure 1.6. Provider market-share trajectory driven by best-in-class ROI.

1.5 Series A Investment: Capital Allocation and Milestones

Arka is raising a **\$25M Series A** to convert a validated, in-production platform into category leadership.

Table 1.10. Series A capital allocation (\$25M).

Use of funds	Allocation	Purpose
Product development & engineering	40% (\$10.0M)	ARKA-CLIN GA; ARKA-ED build; AIIE-OPTIMA O-0→O-5
Sales & go-to-market	35% (\$8.75M)	Regional sales teams; hospital + payer partnerships; clinical advisory board
Data & clinical validation	15% (\$3.75M)	Prospective validation studies; regulatory submissions; evidence generation
Operations & infrastructure	10% (\$2.5M)	Finance, HR, legal, federated-cloud scaling

18-month milestones (post-Series A): deploy ARKA-CLIN in 10 hospital systems representing ~2M annual studies; launch ARKA-ED in 5 major academic centers; begin ARKA-INS pilots with 2–3 major payers; publish 2–3 peer-reviewed validation studies; reach \$2M ARR at >75% gross margin and Rule-of-40 readiness for Series B.

1.6 Go-to-Market: The Three-Wave Approach

Arka's go-to-market deliberately avoids the two classic health-tech traps—the enterprise whale that eats the runway, and the science project with no buyer. It sells to regional and mid-sized groups first (short cycles), runs a payer-first motion where CMS-0057-F is closing the deal, and stages the three products in three waves.

Series A capital allocation

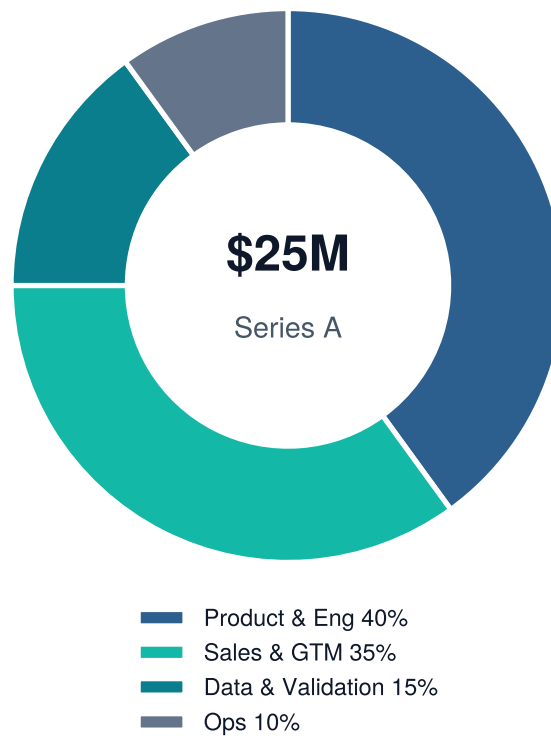


Figure 1.7. Series A capital allocation across product, go-to-market, validation, and operations.

Table 1.11. The three-wave go-to-market.

Phase	Timeline	Focus
Wave 1: ARKA-CLIN	Months 1–18	Early-adopter and regional health systems; case studies; first publications
Wave 2: ARKA-ED	Months 12–30	Academic centers and residency programs; network-effect fly-wheel
Wave 3: ARKA-INS	Months 18–42	Payers and RBMs riding the CMS-0057-F forcing function; 90-day self-funding pilot

1.7 Exit Strategy and Investor Returns

Arka is well-positioned for a 2029–2030 exit. Multiple acquirers—Optum, Change Healthcare, Epic, Cerner/Oracle, and the large RBMs—have strategic incentives to acquire a clinically validated, vendor-neutral, bilateral CDS platform, and the public markets reward regulatory-compliance healthcare software at 12–20× revenue (Figure 1.8).

Table 1.12. Exit scenarios and Series A returns (\$25M invested).

Scenario	Year	Multiple	Valuation	Series A return
Secondary sale to growth equity	2027–28	6–8×	\$1.0–1.4B	40–56×
Strategic acquisition	2028–29	10–12×	\$1.7–2.1B	68–84×
IPO (public markets)	2029–30	15–20×	\$2.6–3.4B	104–136×

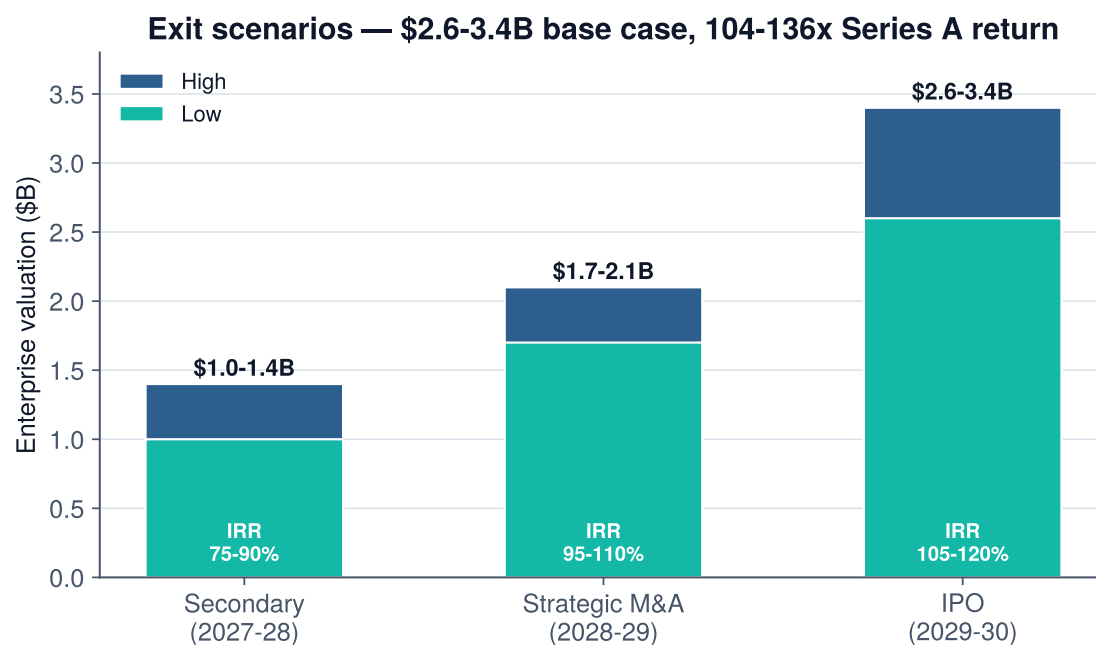


Figure 1.8. Exit scenarios—\$2.6–3.4B base case, 104–136× Series A return.

The ARKA investment opportunity, in one paragraph. Arka is a bilateral clinical decision support platform addressing a \$12.5B direct—and \$30–40B+ total—annual opportunity in imaging appropriateness. It is entering a market at inflection: regulatory mandates are driving adoption, competing solutions are underperforming, and healthcare organizations are actively budgeting for CDS infrastructure. Arka’s three-phase platform diversifies revenue, accelerates adoption through network effects, and is defended by a research-grade engine, a federated data moat, and a Non-Device regulatory posture that lets it ship today. We project **\$172.4M revenue by Year 5 (47% CAGR)** with a **\$2.6B–\$3.4B exit valuation** by 2029–2030, delivering **100–140× returns** for Series A investors. *remARKAbly precise*—at the math, the architecture, the regulatory posture, and the speed of execution.

2 CHAPTER 2

Why ARKA Survives the Health-Tech Graveyard

Most health-tech startups do not fail because their technology does not work. They fail because they built the technology first and went looking for the workflow, the stakeholders, and the money afterward. Arka was built in the opposite order, and the proof is structural rather than aspirational. This chapter explains the failure cycle, the nine specific ways health-tech dies, the pain points Arka addresses, the buyers who feel them, and the competitive matrix—the strategic context an investor needs before underwriting the financial model.

2.1 The Failure Cycle---and the Line ARKA Crosses

The cycle that kills health-tech startups. An innovator spots a problem better technology could fix, assumes nobody has built it, and builds it. They find a few excited health systems, make a few sales, get press, and raise a round. Then implementation hits: clinical workflows are tangled, incentives are misaligned, integrations will not clear IT, and they are learning change management for the first time. The tech *works*—but adoption is low and the outcomes data is thin. The company shuts down, gets acqui-hired, or is sold for parts.

The original sin is the order of operations: the doomed founder builds the tech *first*. The correct order is the inverse—understand the problem, map every stakeholder and their existing workflow, determine the ideal (tech-agnostic) solution, validate the trade-offs with clinicians, and build the tech *last*. Arka was built that way.

Table 2.1. ARKA was built in the correct order—and the proof is structural.

The right step		How ARKA did it (not “plans to”)
Understand the problem		The problem is not “doctors lack a dashboard.” It is that a hospital is paid less than it earned because imaging orders go out without the documentation a payer requires. A money problem, framed in money.
Map every stakeholder		ARKA is bilateral by architecture: the same engine serves the ordering clinician <i>and</i> the payer reviewer. CFO, rev-cycle director, CMIO, ordering physician, prior-auth nurse, and payer medical director are explicit users.
Solve it ideally, tech-agnostic		The ideal solution is “capture the clinical justification once, at the source, in a form both the doctor and payer trust.” A workflow insight, not a tech insight.
Validate trade-offs		ARKA is non-blocking by design—it cannot stop an order, only inform one. The trade-off offered to physicians is <i>zero</i> : no clicks added, full authority retained.
Build the tech last		The engine rides on standards Epic and Cerner already speak. The tech is the easy part precisely because the hard parts were solved first.

The line Arka crosses that the graveyard never reaches: Arka's first sentence to a buyer is about *their money*, and its first promise to a physician is *we will not touch your workflow*. The failed startup's first sentence is about *its technology*, and its first ask of a physician is *change how you work*. That inversion is the whole difference.

2.2 The Nine Ways Health-Tech Dies---and Why ARKA Is the Exception

Table 2.2. The nine failure modes, and why ARKA is the exception to each.

Failure mode	Why ARKA is the exception
1. The reimbursement disconnect	ARKA is built around the billable event—it protects the reimbursement of a high-margin imaging CPT line. Buyer (CFO) and beneficiary (the P&L) are the same.
2. Claim denial ("saving codes, not lives")	Denial recovery is ARKA's core revenue engine: 20–40% imaging denial rates, ~86% avoidable, documented up front so claims go out clean.
3. The regulatory minefield	ARKA was architected inside the Non-Device CDS exclusion from day one—no 510(k); HIPAA handled by never moving raw PHI; CMS-0057-F a tailwind.
4. Integration nightmares	Ships on the rails Epic and Cerner already run: CDS Hooks + SMART-on-FHIR. No proprietary integration to reject.
5. Lack of unmet need ("mansion MVP")	The pain is documented in CAQH, KFF, AMA, ACR, and CMS data—one of the most-quantified problems in US healthcare.
6. The "fancy calculator" trap	Not an LLM wrapper: an explainable rules-first engine upgrading to a research-grade sequential-decision engine. Every output cites a guideline.
7. Baseline drift / alert fatigue	Returns empty cards when no guideline fires; a contextual bandit learns each site and converges from ~15% to ~2% exploration.
8. Stagnant sales cycles	Wedge is the regional/mid-sized group plus a payer-first motion where CMS-0057-F is a January-2027 forcing function.
9. Incumbent blockade & burn-rate burnout	ARKA sits <i>on top</i> of incumbents—a layer over the RBM and a CDS-Hooks app inside the EHR, not a rival to either.

2.3 The Twelve Pain Points

The market problem is not abstract; it is twelve concrete, quantified bleed points that buyers feel every day. Arka addresses every one at the single moment of the order.

2.4 The Buyer-by-Buyer Grid

2.5 Per-Segment Payer Value Cases

The payer opportunity is not monolithic; each segment buys ARKA-INS for a different headline reason.

Table 2.3. The twelve pain points and their quantified scale.

#	Pain point	Scale
1	PA administrative overhead	~\$13B/yr
2	Wrong denials & the appeals tax	80.7% MA overturn
3	Site-of-service leakage	~\$14M/yr per 1M lives
4	Inappropriate / low-value imaging	30–35% of studies
5	Incidental-findings cascade	\$460–\$2,155/finding
6	Fraud, waste & abuse	\$353M identified/yr (RAC)
7	Front-end imaging denials	20–40%
8	Never-reworked write-offs	~50% of denials
9	Throughput backlog (high-margin line)	weeks of delay
10	Prior-auth nurse burnout	~39 PAs, ~13 hrs/MD/wk
11	Provider-payer disagreement loop	\$40B/yr
12	CMS-0057-F non-compliance exposure	8-figure build to avoid

Table 2.4. Every buyer, the pain they own, and the dollar lever ARKA pulls.

Buyer	Pain they own	ARKA dollar lever
Hospital CFO	Imaging revenue leakage	~\$3.5M/yr recovered (mid-sized)
Rev-cycle director	Denials, appeals, write-offs	Clean claim at order; specific PAS reasons
CMIO	EHR integration risk	Standard CDS Hooks app; no proprietary integration
CMO	Appropriateness & quality	Guideline-cited, defensible utilization
CQO	STAR / quality variance	DR-OPE value proof; \$80–120M per STAR point
ACO CFO	Total cost of care	VOI + pathway cost reduction ($\geq 20\%$)
Chief wellness officer	Clinician burnout	Non-blocking, silent-unless-cited
Payer CMO	Medical loss ratio	Inverted denial-risk engine (MLR bps)
Payer CXO/COO	PA admin G&A	35–40% auto-approval; FTE reduction
Payer CIO	CMS-0057-F compliance	Da Vinci PAS endpoint today; 8-figure build avoided

Table 2.5. Per-segment payer value cases.

Segment	Why they buy
Commercial national carriers	Site-of-service leakage and MLR on the imaging line; \$220–350M/yr exposure at 12M lives
Medicare Advantage plans	Appeal-overtake losses and STAR-rating variance; \$80–120M per STAR point on a 500K-life plan
Medicaid MCOs / CHIP	Administrative cost per PA and audit defensibility under tight margins
Tricare / federal carriers	Compliance, auditability, and FWA reduction under federal oversight
RBM (eviCore, Carelon, Cohere)	Per-case margin inverts once 35–40% of orders auto-approve; ARKA is a layer that raises their margin

2.6 The Competitive Matrix

Table 2.6. The competitive matrix—and why ARKA wins each class.

Competitor class	Their position	Why ARKA wins
Imaging-AI SaMD (Aidoc, Viz.ai)	Pixel-level Class II devices; slow, capital-intensive 510(k)s	ARKA is Non-Device, ships today, never analyzes pixels
RBM (eviCore, Carelon)	Labor-intensive retrospective review	ARKA is a layer on top that raises their per-case margin
PA-UX startup (Cohere Health)	UX-led, manual clinical review	ARKA is the only Da Vinci-native engine with SHAP-explained reasoning
EHR-native CDS (Epic alerts)	Generic, not appropriateness-grade, payer-blind	ARKA is bilateral, guideline-cited, and explainable
Foundation-model entrant	Cannot satisfy the independent-review criterion; cannot buy the federated dataset	ARKA's explainability + federated outcome data are unreplicable

2.7 The Moat: Six Reasons It Cannot Be Copied

1. **The bilateral engine.** The only engine running on both sides of the prior-auth wall, from the same math. No payer can build it without being the provider-side CDS vendor; no provider tool without being the payer-side engine. Arka occupies both.
2. **The federated data moat.** Every new site improves the engine for all sites—without anyone shipping a patient record. A labeled, payer-adjudicated, multi-institution outcome dataset money cannot buy and a foundation model cannot synthesize.
3. **Non-Device clearance to ship today.** While a SaMD competitor spends two years and seven figures per indication on 510(k)s, Arka deploys into any CDS-Hooks EHR now. Speed and capital efficiency are the moat.
4. **Standards-native integration.** CDS Hooks + SMART-on-FHIR + Da Vinci clears IT review as a standard app, not a proprietary risk.

5. **Continuous statistical proof of value (DR-OPE).** A renewal conversation where the dashboard already shows dollars recovered is not a competitive evaluation. The anti-churn weapon and the fundraising weapon at once.
6. **Imaging is just the wedge.** The same sequential-decision engine will, in 3–5 years, reason over specialty-drug step therapy, elective-surgery appropriateness, and any expensive medical decision against the evidence—the \$10B+ appropriateness layer of American medicine.

2.8 What Has to Be True

A disciplined investor underwrites the assumptions, not the conclusions. Arka's thesis rests on five propositions, each of which is either already demonstrated or conservatively bounded:

1. **Hospitals will pay to recover denied imaging revenue.** Demonstrated: denial recovery maps to an existing budget line (rev-cycle) and an existing buyer (the CFO), and the floor model requires no behavior change—only collecting earned revenue.
2. **ARKA can integrate without a multi-quarter IT fight.** Bounded: it ships on CDS Hooks + SMART-on-FHIR + Da Vinci, the same rails Epic and Cerner already run; there is no proprietary integration to reject.
3. **The Non-Device posture holds.** Controlled: the four criteria are satisfied at the code level and enforced by a CI boundary lint; the one feature that would break it (Epoch 2) is excluded.
4. **Physicians will not disable it.** Designed: non-blocking, silent-unless-cited, sub-800ms, with a per-site bandit and a clinical-champion program—the direct antidotes to the alert-fatigue death.
5. **The payer market opens on schedule.** Externally forced: CMS-0057-F mandates the FHIR PA rail by January 2027 regardless of ARKA, and ARKA is the cheapest compliant path.

If these five hold—and the first four are within Arka's control—the financial model follows. The thesis does not require a scientific breakthrough or a behavior change; it requires execution against a problem the market has already quantified and a mandate the regulator has already written.

The strategic thesis. Arka is the exception to all nine failure modes not by luck but by design: it leads with the buyer's money, never touches the doctor's workflow, was born inside the regulatory exclusion, ships on the rails the industry already runs, and is defended by a bilateral position and a federated dataset that no single-sided competitor can assemble. The financial model in Chapter 5 rests on this strategic foundation.

The ARKA Imaging Intelligence Engine

The ARKA Imaging Intelligence Engine (AIIE) represents a fundamental reimagining of clinical decision support for medical imaging. Rather than relying on static, expert-consensus criteria that have become functionally obsolete in modern healthcare, AIIE deploys a dynamic, quantitatively rigorous, and transparently reasoned system that delivers appropriate imaging recommendations at the moment of clinical decision-making—and, critically, the *same* engine that advises the ordering clinician scores the order's denial risk for the payer. AIIE is the single technological core beneath ARKA-CLIN, ARKA-ED, and ARKA-INS.

The core problem. The American College of Radiology (ACR) Appropriateness Criteria—once the gold standard—has become structurally compromised. In 2024, the ACR granted an exclusive, perpetual license to Optum, consolidating control of imaging-appropriateness standards in a single for-profit corporation with direct conflicts of interest. The criteria themselves remain consensus-based, updated infrequently, and employ reasoning that lacks transparency and quantitative rigor. Override rates of 49–96% across health systems demonstrate that clinicians do not trust the current approach.

The ARKA solution. AIIE is purpose-built under the FDA's Non-Device Clinical Decision Support framework established by Section 3060 of the 21st Century Cures Act. It operates as a learning system that continuously integrates clinical evidence, patient-specific factors, institutional context, and economic considerations. AIIE's reasoning is fully explainable at every decision point, deployed across three integrated platforms, and optimized for regulatory compliance, clinical adoption, and superior patient outcomes.

3.1 What Is ARKA AIIE?

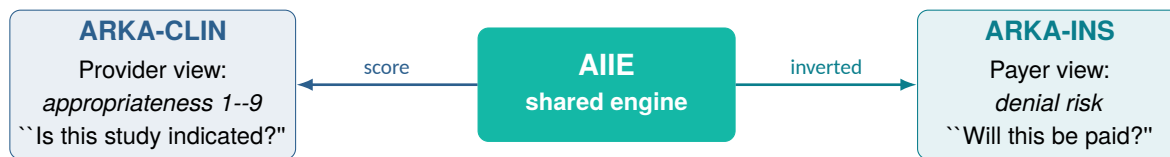
AIIE is a hybrid, evidence-weighted reasoning engine that takes structured clinical context as input and returns a quantitative appropriateness judgment, a denial-risk routing decision, a pre-filled documentation packet, and a human-readable explanation—all in under 800 milliseconds. It is the shared brain between the provider-facing and payer-facing products.

3.1.1 The Bilateral Engine: One Math, Both Sides of the Wall

The structural insight that no competitor holds is that the appropriateness score a clinician needs and the denial-risk score a payer needs are the *same computation, read in two directions*. AIIE computes a RAND/UCLA appropriateness score and inverts it into a denial-risk signal; the provider's view and the payer's view are therefore guaranteed to be consistent. When both sides reason from identical math with identical citations, there is no inconsistency to litigate—which is exactly why appeal-overturn rates collapse.

Table 3.1. The three AIIE-powered platforms.

Platform	Core function	Integration points
ARKA-CLIN	Real-time appropriateness assessment at point of order; clinical justification capture	EHR order entry, radiology workflow, clinical documentation
ARKA-ED	Realistic case simulation and feedback that trains clinicians on appropriateness	LMS integration, residency curricula, CME platforms
ARKA-INS	Prospective and concurrent review with evidence-based authorization; retrospective analytics	Claims systems, member-data platforms, Da Vinci PAS workflows

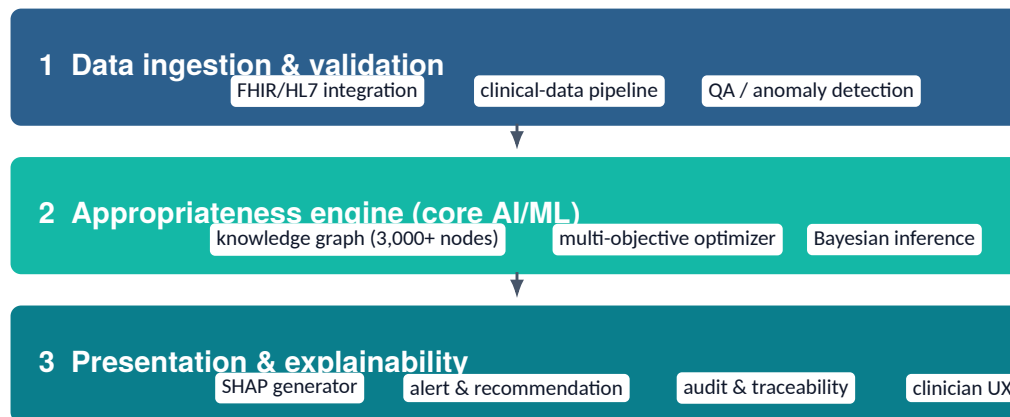


*Identical evidence base · identical citations · identical objective weights
 ⇒ no inconsistency to litigate ⇒ appeals evaporate*

Figure 3.1. The bilateral engine—the position no payer or provider tool can occupy alone.

3.1.2 Core Technical Architecture: A Three-Layer Design

AIIE employs a three-layer technical architecture that cleanly separates data ingestion, appropriateness reasoning, and clinical presentation. This modular design ensures that each layer can be independently validated, updated, and audited.

**Figure 3.2.** AIIE's three-layer architecture: ingestion, reasoning, and explainable presentation.

3.2 AIIE Data Flow: From Patient to Recommendation

When a clinician initiates an imaging order—or when a payer performs review—AIIE executes a deterministic, auditable flow in five steps (Figure 3.3).

Table 3.2. The three architectural layers in detail.

Layer	Purpose	Key technologies
Data ingestion & validation	Extract and validate patient data from the EHR; normalize across heterogeneous systems; detect anomalies	HL7/FHIR interfaces, validation frameworks, anomaly detection
Appropriateness engine	Encode the evidence hierarchy; compute appropriateness scores; generate confidence intervals; optimize competing objectives	Knowledge-graph representation, Bayesian inference (PyMC/Stan), Pareto optimization, constraint solvers
Presentation & explainability	Generate human-readable explanations; present context-aware alerts; maintain a complete audit trail	SHAP/TreeSHAP, natural-language generation, workflow integration, audit logging

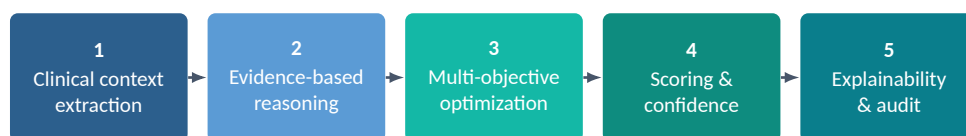


Figure 3.3. The deterministic, auditable AIIE decision flow.

Step 1—Clinical context extraction. AIIE integrates with the EHR to extract demographics, clinical history, current findings, prior imaging, radiation-exposure history, relevant labs, comorbidities, and contextual factors (location, resource availability, urgency). All extracted data is validated against quality thresholds and flagged for manual review when completeness or validity concerns arise.

Step 2—Evidence-based reasoning. The Clinical Knowledge Graph—more than 3,000 evidence nodes—is queried. Each node is weighted by evidence hierarchy (RCTs > high-quality observational studies > expert consensus), and temporal-decay functions ensure older evidence gradually loses weight as newer evidence accumulates.

Step 3—Multi-objective optimization. Rather than producing a single score, AIIE formulates the decision as a constrained multi-objective problem—maximizing diagnostic yield while minimizing radiation, cost, patient burden, and resource use—and produces a Pareto-optimal frontier of options.

Step 4—Scoring and confidence. AIIE generates a primary appropriateness score (1–9) with a 95% Bayesian credible interval. Communicating genuine uncertainty, rather than false certainty, is essential to clinician trust and regulatory defensibility.

Step 5—Explainability and audit. For every recommendation, a SHAP waterfall decomposition generates a human-readable explanation of which factors most strongly influenced the score—auditable, traceable, and sufficient for both clinician review and regulatory inspection.

3.2.1 Latency Engineering: How the 800ms Budget Is Met

The two-tier design is what makes sub-800ms explainable inference possible. Tier one is a deterministic rules engine that resolves the 60–70% of orders with an unambiguous guideline match in under 5ms—a recent-prior redundancy, a clear red flag, a documented conservative-management gap. Only the residual complex cases invoke tier two, the ML inference path, which is itself optimized: the heterogeneous graph encoder produces a compact 128-dimension context vector rather than invoking a large transformer, MCTS roll-outs are capped to fit the latency budget, and results are cached at the order-context level. Critically, the explanation is not computed after the fact—SHAP attributions are produced alongside the score, so the card returns complete. If any model call exceeds its time budget,

a circuit-breaker returns the tier-one rules result tagged `fallback=true`, so the clinician never waits. The result is a system that is simultaneously fast, explainable, and fail-safe.

3.3 The Clinical Knowledge Graph: The Reasoning Core

At the heart of AIIE lies a structured Clinical Knowledge Graph comprising over 3,000 evidence nodes. This is neither a brittle rule base nor an opaque neural network; it is a meticulously curated, evidence-weighted network that encodes clinical knowledge in a form that is simultaneously rigorous and explainable.

Node types. The graph includes nodes for clinical presentations (symptoms, findings, diagnoses), imaging modalities (CT, MRI, ultrasound, nuclear medicine, etc.), clinical outcomes (diagnostic accuracy, therapeutic impact, adverse events), evidence sources (individual RCTs, meta-analyses, observational cohorts, expert consensus), and contextual factors (demographics, institutional capability, cost structures). Edges represent causal relationships, evidence associations, and conditional dependencies.

Evidence weighting. Each edge carries a weight reflecting its evidentiary support, assigned per the GRADE framework—RCTs above observational studies, observational above expert consensus—so stronger evidence naturally dominates weaker evidence in appropriateness calculations.

Temporal dynamics. Clinical evidence evolves continuously. AIIE incorporates temporal-decay functions so that a meta-analysis from 2010 gradually loses relative influence as new RCTs are published in 2024–2026, keeping recommendations current without manual re-curation for every update.

Structural validation. The graph is validated for logical consistency, circularity detection, and semantic coherence. Automated tools identify orphaned nodes, conflicting relationships, and temporal inconsistencies; all findings are reviewed by clinical content experts before deployment.

Table 3.3. AIIE input and output specifications.

Category	Items	Status
IN: Patient demographics	Age, gender, BMI, pregnancy status, renal function	Required
IN: Clinical presentation	Chief complaint, pertinent history, physical findings, labs	Required
IN: Imaging history	Prior studies, dates, results, prior procedures	Required
IN: Radiation exposure	Cumulative dose, age when exposed, imaging intervals	Optional
OUT: Appropriateness score	Primary score (1–9), credible interval, confidence	Always
OUT: Clinical guidance	Recommended modality, alternatives, timing	Always
OUT: Explanation	SHAP waterfall, key factors, evidence summary	Always
OUT: Denial-risk routing	AUTO_APPROVE / LIKELY_APPROVE / CLINICAL_REVIEW / LIKELY_DENY	Always

3.4 AIIE Scoring Methodology: Quantitative Rigor and Transparency

AIIE’s deterministic v1 core—always on, and the fallback whenever the ML layer is unavailable—computes the appropriateness score as a transparent set of bounded, GRADE-aligned deltas from a neutral baseline of 5 on the RAND/UCLA 1–9 scale. There are no hidden coefficients: every factor’s weight is fixed, published, and auditable (Figure 3.4).

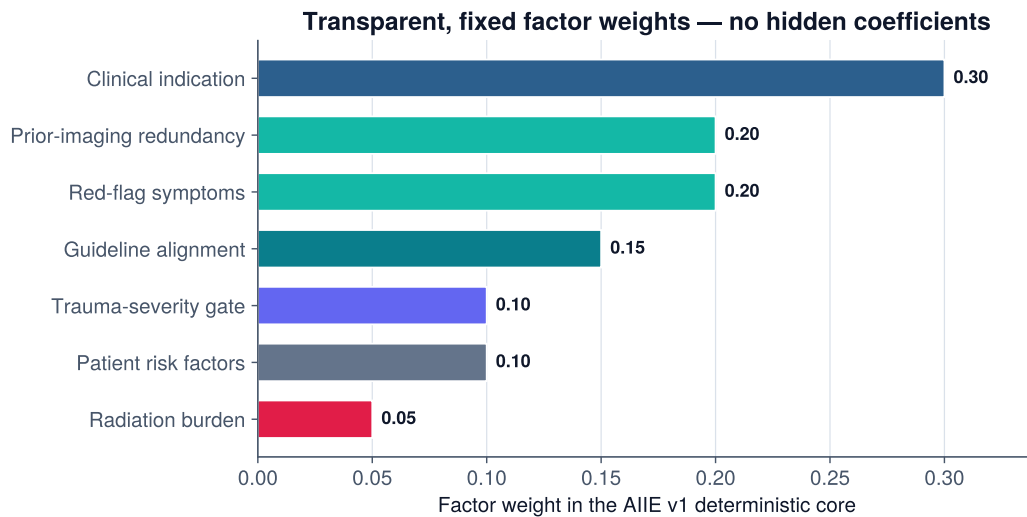


Figure 3.4. The fixed factor weights of the AIIE v1 deterministic core.

Table 3.4. AIIE scoring factors, weights, and evidentiary basis (deterministic core).

Factor	Weight	Evidence basis
Clinical indication strength	0.30	GRADE summary-of-findings for symptom–test concordance; RAND/UCLA indication panels
Prior-imaging redundancy	0.20	Choosing Wisely / ACR criteria on redundant imaging in stable presentations
Red-flag symptoms	0.20	ACEP and specialty red-flag guidance for urgent diagnoses
Guideline alignment	0.15	GRADE evidence-to-decision tables favoring conservative care first when appropriate
Trauma-severity gate	0.10	AAST ISS/AIS and ACR head-trauma criteria (GCS, REMS modifiers)
Patient risk factors	0.10	Population risk modifiers (age, immunosuppression, pregnancy)
Radiation-exposure burden	0.05	BEIR / ICRP-informed trade-offs for ionizing modalities

3.4.1 Detailed Scoring-Factor Breakdown

Each factor is more than a weight—it encodes a specific clinical reasoning pattern grounded in published evidence. Table 3.5 explains how each factor works and its directional impact on the appropriateness score.

Each factor emits a bounded contribution; the contributions sum onto the baseline and the result is clamped to the 1–9 range. The score is then inverted into a denial-risk proxy, paired with the order’s CPT and ICD-10 context to compute a medical-necessity (MNAI) tier, and accompanied by a confidence value derived from the completeness of the supplied context. The result is a fully decomposable score: the SHAP waterfall in Figure 3.5 shows exactly how a representative order reaches 7.2 (Usually Appropriate) with a 95% credible interval of 5.8–8.4.

Why this matters for the FDA boundary. Because the primary basis of every recommendation is a *published guideline cited above any ARKA score*, and because the deterministic core can render the entire decision without the ML layer, AIIE satisfies the “independent review” criterion of Non-Device CDS by construction. The ML layer only *refines* a recommendation a guideline already supports—it never originates one.

Table 3.5. How each AIIE factor works and shapes the score.

Factor	How it works and its impact
Clinical indication strength	Matches the presentation against the knowledge graph to identify relevant diagnostic hypotheses. Strong positive when the indication is well-defined and matches an evidence-based pathway; negative when vague or non-standard.
Prior-imaging redundancy	Checks temporal spacing and relevance of priors. Strong negative if recent, relevant imaging exists; positive if clinical status has materially changed since the prior study.
Red-flag symptoms	Elevates appropriateness for urgent presentations (neurologic deficit, bladder/bowel dysfunction, cancer history). Strong positive when high-acuity red flags are present.
Guideline alignment	Rewards documented conservative management before advanced imaging when guidelines call for it (e.g., 4–6 weeks of physical therapy for low-back pain). Negative when first-line care was skipped.
Patient risk factors	Adjusts for age extremes, pregnancy, renal dysfunction, and immunosuppression—shifting away from high-dose or contrast modalities where indicated.
Radiation-exposure burden	Penalizes ionizing modalities (CT, PET, nuclear) relative to ultrasound and MRI when diagnostic yield is comparable; stricter limits for pediatric and pregnant patients (ALARA).
Trauma-severity gate	For trauma orders, incorporates structured injury-severity scores (ISS/AIS, GCS) so that genuinely high-acuity presentations are not down-scored.

AIIE SHAP waterfall — every score decomposed into cited factors (95% CI 5.8-8.4)**Figure 3.5.** SHAP waterfall—a representative order decomposed into cited factor contributions.

3.4.2 A Worked Clinical Example

Consider a representative order to make the flow concrete. A 38-year-old with six weeks of low-back pain, no neurologic deficit, no red flags, and no documented conservative management; the clinician selects “MRI lumbar spine.” AIIE extracts the structured context, queries the knowledge graph, and computes:

- **Guideline alignment** fires *negative*: GRADE evidence-to-decision tables and Choosing Wisely recommend 4–6 weeks of conservative management (physical therapy, NSAIDs) before advanced imaging for uncomplicated low-back pain. No conservative care is documented.
- **Red-flag symptoms** contribute nothing: none are present.
- **Prior-imaging redundancy** is neutral: no recent prior.
- The order scores **3.4/9 (Rarely Appropriate)**, which inverts to a LIKELY_DENY routing.

Rather than blocking the order, AIIE returns a non-modal card: the 3.4 score with its SHAP factors and citations, a recommended alternative (“conservative management × 6 weeks, then re-evaluate”), and—if the clinician proceeds—a pre-filled DTR Questionnaire asking them to document the one factor that would justify the study (e.g., a red flag not captured in the chart). If a genuine red flag exists, the clinician documents it, the score rises, and the packet goes out *complete and defensible*. If not, the low-value study is averted at the source. Either way, the physician retains full authority and the payer receives a cited, appeal-proof rationale.

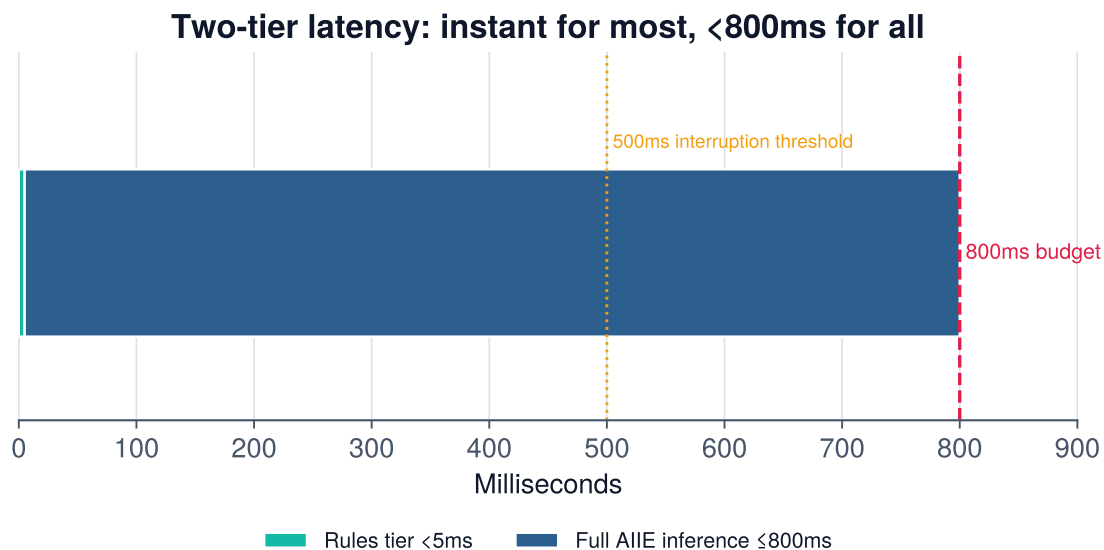


Figure 3.6. The two-tier latency budget—instant for most cases, under 800ms for all.

This entire exchange completes within the latency budget shown in Figure 3.6: the rules tier resolves the guideline conflict in under 5ms, and the full explainable card—score, factors, citations, alternative, and pre-filled questionnaire—returns in under 800ms, well inside the 500ms-plus threshold at which clinicians begin to perceive an interruption.

3.4.3 Confidence and the Medical-Necessity Tier

Two further outputs accompany every score. First, a **confidence value** is computed from the completeness of the supplied context—the presence of a detailed chief complaint, documented prior-imaging timeframe, conservative-management duration, the order’s CPT, and the payer—rising as the record is more complete and capping below

certainty to preserve appropriate humility. Second, for orders carrying a CPT and ICD-10 context, AIIE computes a **medical-necessity (MNAI) tier**—a green/amber/red classification of how defensible the order is against coverage policy. The MNAI tier is what drives the payer-side routing and the DTR documentation prompt: a “green” order needs nothing further, while an “amber” order surfaces exactly the one questionnaire item that would move it to green. This is how the engine converts a continuous appropriateness score into the discrete, actionable artifacts that both the clinician and the payer consume.

3.5 Multi-Objective Optimization: Navigating Trade-Offs

Traditional appropriateness systems collapse decision-making into a single score. AIIE recognizes that real clinical decisions involve trade-offs among competing objectives, and makes them explicit across five dimensions: (1) diagnostic yield, (2) radiation-exposure minimization, (3) cost-effectiveness, (4) patient convenience and burden, and (5) resource-utilization efficiency. Rather than imposing arbitrary weights, AIIE identifies the Pareto-optimal frontier—the set of recommendations such that no alternative strictly dominates them across all five objectives—and presents multiple options along that frontier (Figure 3.7).

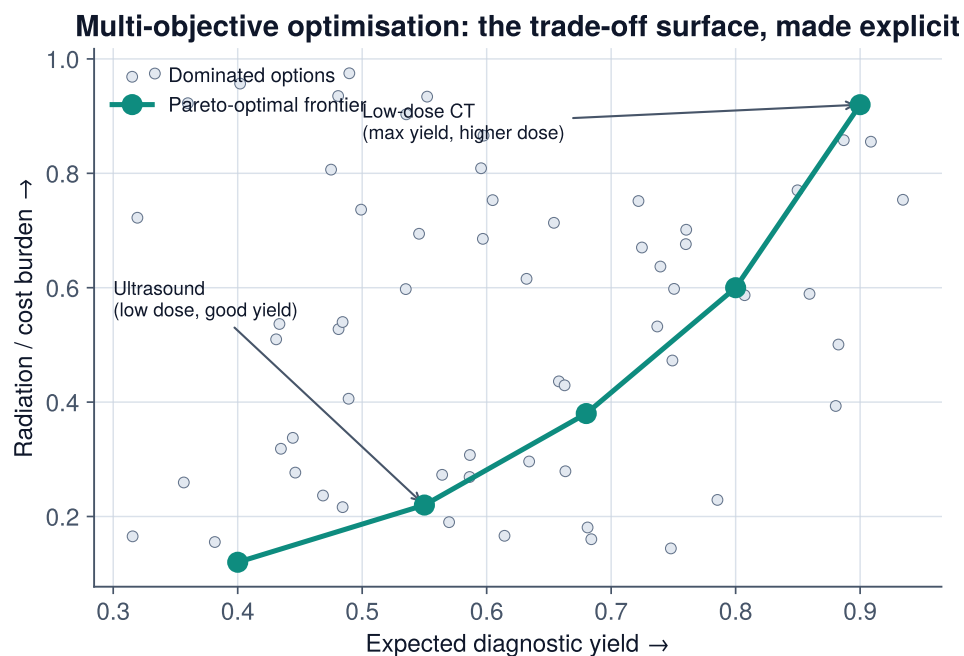


Figure 3.7. The multi-objective trade-off surface, made explicit rather than hidden.

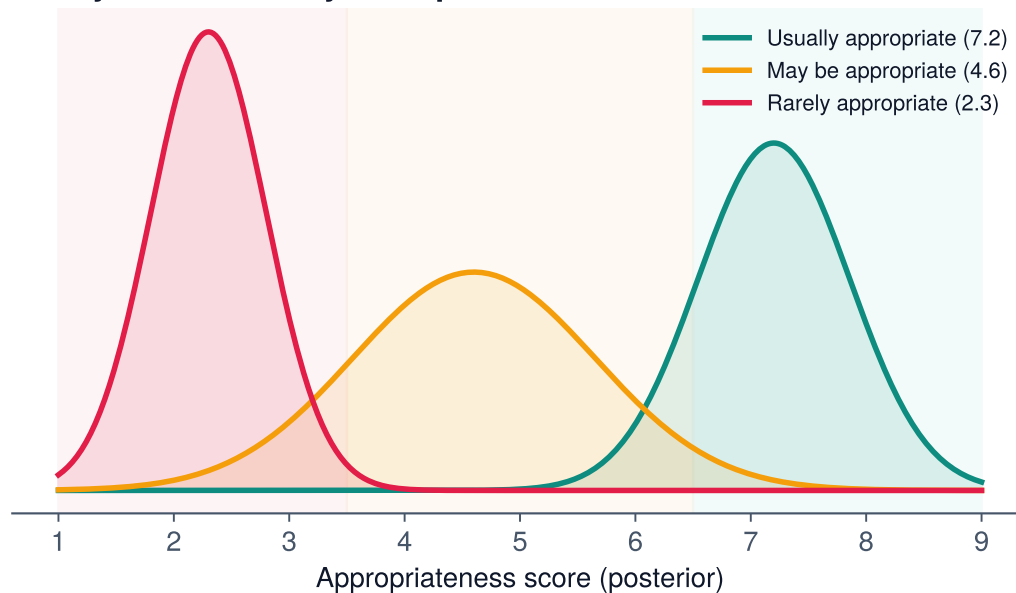
3.6 The 1--9 Scale, GRADE, and Bayesian Uncertainty

AIIE adopts the familiar 1–9 scale to minimize relearning, but its interpretation differs fundamentally: each score is evidence-based, quantitatively derived, and accompanied by a credible interval. Evidence is weighted via the **GRADE** framework (Level A High through Level D Very Low), the international standard used by the WHO and Cochrane. And every score is modeled as a Bayesian posterior: a recommendation might read “Primary score: 7.2 (95% CI 5.8–8.4),” communicating both the central estimate and the genuine state of knowledge (Figure 3.8). Sources of uncertainty—evidence heterogeneity, missing clinical parameters, extrapolation across populations, and model uncertainty—are represented explicitly.

Table 3.6. The AIIE 1–9 appropriateness scale and its interpretation.

Score	Category	Interpretation
1–3	Rarely appropriate	Imaging unlikely to yield useful information or poses unjustifiable risk/cost relative to benefit; alternatives strongly preferred
4–6	May be appropriate	Reasonable likelihood of useful information but significant uncertainty; clinical judgment and patient context should drive the decision
7–9	Usually/always appropriate	Strongly indicated; high diagnostic yield and/or therapeutic-impact likelihood; risk/cost/burden justified

Every score is a Bayesian posterior with a 95% credible interval

**Figure 3.8.** Every appropriateness score is a Bayesian posterior with a 95% credible interval.

3.7 AIIE-OPTIMA: The Next-Generation Engine

The deterministic v1 stack (RAND/UCLA + GRADE + gradient-boosted ML + SHAP) is in production today and remains the always-on fallback. Behind it, ARKA is rolling out **AIIE-OPTIMA**, a research-grade sequential-decision engine that upgrades AIIE from “score the single study” to “reason over the whole imaging pathway”—without changing a single clinician-facing screen.

The quant-finance pedigree, made literal. AIIE-OPTIMA is a partially observable Markov decision process (POMDP) solved with a Bayesian Value-of-Information kernel, AlphaZero-style Monte-Carlo tree search over imaging *pathways*, a constrained Thompson-sampling bandit, a heterogeneous graph-neural-network encoder, federated learning with differential privacy, and doubly-robust off-policy evaluation. Every layer is explainable, and every layer reasons over structured facts—never pixels. It reasons the way physicians actually think: “ultrasound now → MRI only if indeterminate.”

The POMDP formalism. OPTIMA models the imaging decision as a partially observable Markov decision process: a hidden disease state s , a belief $b(s)$ maintained over $\sim 1,500$ ICD-10-mapped labels, actions a (image now, image later, choose modality, defer), and a multi-objective reward $r = \alpha \cdot \Delta\text{Info} - \beta \cdot \text{dose} - \gamma \cdot \text{cost} - \delta \cdot \text{time} - \epsilon \cdot \text{harm} + \zeta \cdot \text{patientPref}$. The Value of Information of a study is the expected reduction in the entropy of $b(s)$; MCTS searches the tree of imaging *pathways* to maximize discounted reward subject to radiation and cost constraints; and the policy is re-validated weekly by doubly-robust off-policy evaluation. Every term is observable, every weight is surfaced, and every recommendation exposes its Q -value provenance—so the most mathematically sophisticated engine in imaging CDS is also the most explainable.

3.7.1 The Seven Modules

Table 3.7. AIIE-OPTIMA's seven modules—each a structural advantage no rule engine can replicate.

Module	What it does
1. Bayesian VOI kernel	Maintains a belief over $\sim 1,500$ ICD-10-mapped condition labels; rewards diagnostic-entropy reduction (Value of Information), observable at order time
2. Trajectory simulator (Neural ODE + SCM)	Well-defined counterfactuals via do-calculus; fenced to imaging-pathway utility only—never a time-critical event prediction
3. MCTS over pathways (PUCT)	20–80 roll-outs per decision inside the latency budget; ranks pathways with radiation/cost safety constraints
4. Constrained contextual bandit (Thompson)	Chooses which top- k option to surface per site/payer; converges from $\sim 15\%$ to $\sim 2\%$ exploration; cuts alert fatigue
5. Heterogeneous GNN encoder	A relational graph attention network over a curated clinical knowledge graph yields a 128-dimension context vector
6. Federated learning + differential privacy	FedAvg/FedProx, secure aggregation, per-institution ϵ -budget ledger; no raw record ever moves
7. Doubly-robust off-policy evaluation	Re-evaluates each deployed policy weekly versus the trailing 30 days; a $>3\%$ regression auto-rolls-back

3.7.2 The OPTIMA Differentiators (D19--D24)

The headline number that fronts every payer and ACO pitch: a pre-registered validation target of $\geq 15\%$ cumulative radiation reduction and $\geq 20\%$ cost reduction at non-inferior diagnostic yield (Figure 3.10).

3.7.3 The Differentiators in Detail

Each OPTIMA differentiator attacks a specific, quantified cost pool:

- **D19 — The Value-of-Information card.** Shows the expected diagnostic information each study yields (“ultrasound resolves ~ 0.82 of the uncertainty; MRI ~ 0.86 but $+7$ days and $4\times$ cost”). Every avoided low-yield study

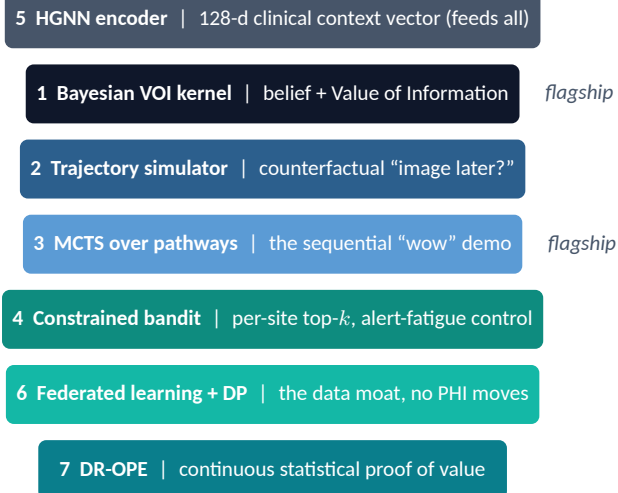


Figure 3.9. The AIIE-OPTIMA module stack—all explainable, all Non-Device.

Table 3.8. The OPTIMA differentiators and the cost pools they attack.

ID	Differentiator	Pool attacked
D19	Value-of-Information card—shows expected diagnostic information each study yields	\$12–30B low-value imaging
D20	Sequential pathway co-pilot (flagship)—ranked pathways with Q -values	Radiologist-throughput crisis
D21	Federated OPTIMA network—per-institution ϵ -budget ledger	The data moat
D22	Continuous value proof / DR-OPE—weekly re-scoring, auto-rollback	\$80–120M per MA STAR point
D23	Auditable multi-objective weights—same weights for provider and payer	\$40B provider–payer disagreement
D24	Cold-start warm-start—federated priors shorten time-to-value	GTM / sales-cycle acceleration

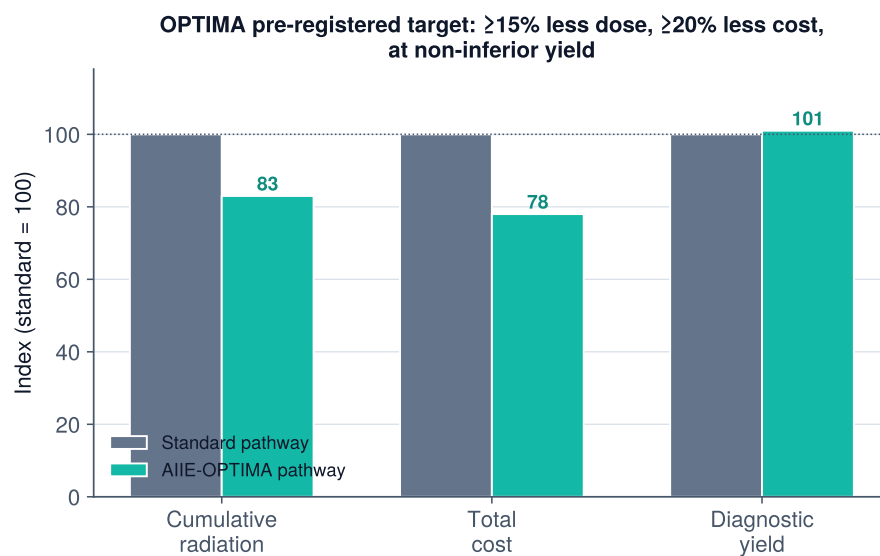


Figure 3.10. OPTIMA's pre-registered validation target.

is worth \$400–\$2,155; at an ACO running 300K studies/yr, a single-digit-percent shift is a multi-million-dollar swing against the \$12–30B low-value-imaging pool.

- **D20 — The sequential pathway co-pilot (flagship).** Surfaces ranked imaging *pathways* with Q -values, attacking the radiologist-throughput crisis (utilization is projected to rise 16.9–26.9% by 2055), wrong-modality-first ordering, and the avoidable-escalation tax. The headline validation target: $\geq 15\%$ cumulative radiation reduction and $\geq 20\%$ cost reduction at non-inferior yield.
- **D21 — The federated OPTIMA network (data moat).** Federated learning of the full belief and value models, with a per-institution ε -budget ledger. Every new site improves all sites without shipping a record.
- **D22 — Continuous value proof / DR-OPE.** Weekly re-scoring against the trailing 30 days, with $>3\%$ regressions auto-rolled-back. A single Medicare-Advantage STAR point on a 500K-life plan is worth \$80–120M/yr—exactly the variance a chief quality officer is measured on.
- **D23 — Auditable multi-objective weights.** The objective $\alpha \cdot \Delta \text{Info} - \beta \cdot \text{dose} - \gamma \cdot \text{cost} - \delta \cdot \text{time} - \epsilon \cdot \text{harm} + \zeta \cdot \text{patientPref}$ uses learned, surfaced per-institution and per-payer weights, attacking the \$40B/yr provider-payer disagreement loop by showing both sides the same math.
- **D24 — Cold-start warm-start.** A new site boots on federated warm-start plus data-free epidemiologic priors and runs a 30-day shadow before surfacing anything—shortening sales cycles and time-to-value.

3.7.4 Discipline: Epoch 1 In, Epoch 2 Out

The boundary that makes OPTIMA fundable. Every gain above is achieved without analyzing a pixel, predicting a time-critical event, taking an autonomous action, or filing a 510(k). **OPTIMA “Epoch 2”** (autonomous triage of low-risk presentations) is *deliberately excluded*: it would convert ARKA from a fast, capital-efficient Non-Device company into a slow, capital-intensive Class II SaMD company competing on a device-maker’s regulatory turf. Take Epoch 1 in full; leave Epoch 2 on the shelf. *The discipline of staying Non-Device is not a limitation on the strategy—it is the strategy.*

The engine rolls out behind a single feature flag, shadow $\rightarrow$ mirror $\rightarrow$ opt-in A/B $\rightarrow$ default-with-override, with any timeout falling back to the v1 deterministic core (tagged `fallback=true`). The full O-0 $\rightarrow$ O-5 execution sequence appears in Chapter 5 and Chapter 8.

3.8 The Federated Network and the Data Moat

The most durable consequence of the AIIE-OPTIMA architecture is the federated network it creates. Every institution that deploys Arka contributes to a shared, continuously improving model *without ever transmitting a patient record*: models train locally, only differentially private gradient aggregates are exchanged, secure aggregation prevents the coordinator from seeing any single site’s contribution, and a per-institution privacy-budget ledger bounds cumulative disclosure. The result is an asset that compounds with scale and that no competitor can buy: a labeled, payer-adjudicated, multi-institution outcome dataset spanning urban academic centers and rural critical-access hospitals alike.

3.9 AIIE versus the ACR Appropriateness Criteria

For decades the ACR Appropriateness Criteria served as the de facto standard. Structural changes have undermined that role, and AIIE is designed for the post-ACR environment. Across seven critical dimensions—evidence rigor,

Table 3.9. Why the federated network is an unreplicable moat.

Property	Consequence
Compounding accuracy	Each new site improves calibration for all sites; the model a late entrant would need does not exist to be bought
Privacy-preserving by construction	DP + secure aggregation means the data advantage is achieved without the HIPAA exposure of a central lake
Payer-adjudicated labels	Outcomes are tied to real authorization and payment decisions—ground truth a synthetic dataset cannot fabricate
Cold-start defense (warm-start)	New sites boot on federated priors and run a 30-day shadow, shortening time-to-value and sales cycles

transparency, patient specificity, evidence currency, explainability, clinical integration, and innovation—AIIE's profile consistently extends further from the center than ACR's (Figure 3.11).

AIIE vs. ACR — multi-dimensional performance profile

**Figure 3.11.** AIIE versus ACR—a multi-dimensional performance profile.

Table 3.10. Methodology and reasoning: AIIE versus ACR.

Dimension	AIIE	ACR
Evidence base	Multi-source synthesis, formally GRADE-graded and quantitatively weighted	Expert consensus; opaque, panel-composition-dependent
Reasoning model	Quantitative multi-objective optimization with an explicit Pareto frontier	Qualitative consensus judgment
Score basis	Evidence-based posterior mean with credible intervals	Panel majority vote on category boundaries
Explainability	SHAP waterfall explains every score at line-item level	Black-box consensus; rationale rarely visible
Evidence currency	Continuous updates with temporal decay	Updated every 3–5 years via expert re-convening

Table 3.11. Clinical utility, technology, and business positioning: AIIE versus ACR.

Dimension	AIIE	ACR
Alert design	Context-aware, in-workflow, with actionable alternatives	Reference material; manual lookup
Override rates	Expected 8–25%; overrides logged for feedback	Observed 49–96%
Radiation tracking	Integrated cumulative-dose tracking (ALARA)	None
EHR integration	Deep bi-directional via CDS Hooks / FHIR	Documentation-level only
AI architecture	Hybrid symbolic + ML; explainable core	Static if-then rule engine
Uncertainty	Explicit Bayesian credible intervals	Presented as certain
Licensing	Vendor-neutral; ARKA retains IP	Exclusive perpetual license to Optum

3.9.1 The ACR Problem: Structural Obsolescence

Optum is simultaneously America's largest health plan, a pharmacy-benefit manager, and a utilization-review provider that profits directly from imaging denials. Granting it exclusive control of imaging standards is equivalent to letting a bank set interest-rate standards. Meanwhile, despite two decades as the "gold standard," ACR adoption remains below 10% of health systems for actual clinical decision support; the criteria are static between 3–5-year panels; the consensus process is opaque; override rates run 49–96%; and there is no patient specificity or cumulative-radiation tracking. AIIE answers every one of these limitations. The most telling signal is the override rate—the fraction of recommendations clinicians ignore. ACR's 49–96% override rate is among the highest of any CDS system, while AIIE's evidence-cited, in-workflow design is expected to hold overrides to 8–12% (Figure 3.12). Trust, measured this way, is the difference between a tool that is used and one that is disabled—and it is why AIIE's adoption curve escapes the single-digit ceiling that has constrained ACR for fifteen years (Figure 3.13).

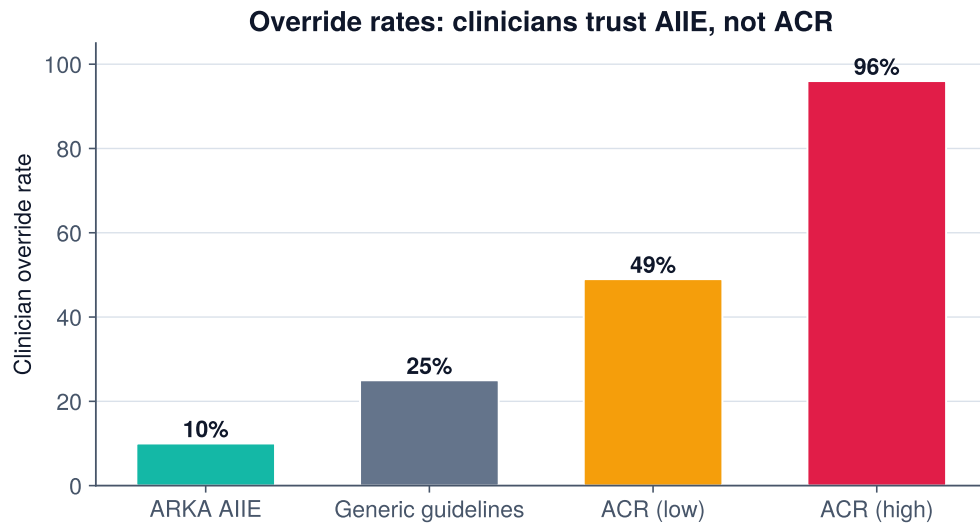


Figure 3.12. Override rates—clinicians trust AIIE's cited, in-workflow recommendations.

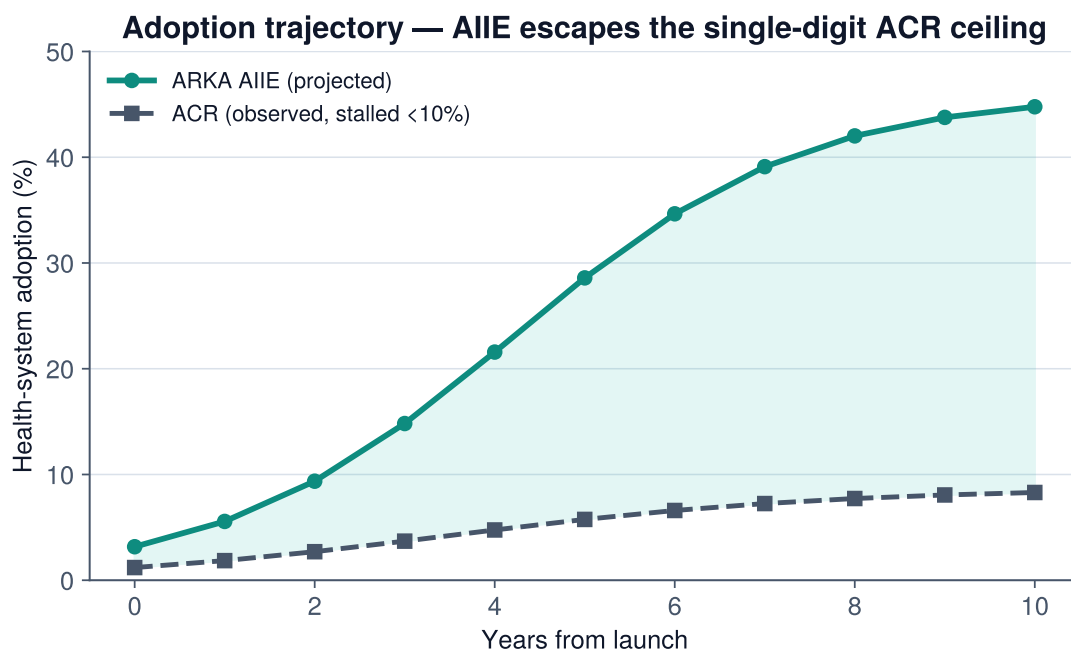


Figure 3.13. Projected adoption—AIIE escapes the single-digit ACR ceiling.

4 CHAPTER 4

FDA Non-Device CDS Regulatory Strategy

Arka's regulatory posture is the chassis the product was built on, not a hurdle it cleared. Four bodies of regulation matter—FDA medical-device law, HIPAA, the CMS interoperability mandate, and state gold-card and appeal statutes—and Arka is on the right side of all four *by construction*. This chapter explains why the Non-Device Clinical Decision Support pathway is the strategy itself, not a constraint upon it.

Binding posture. Arka is, and is architected to remain, a **Non-Device Clinical Decision Support** function under **§520(o)(1)(E) of the FD&C Act** (21st Century Cures Act §3060), as clarified by FDA's CDS Software Final Guidance of January 6, 2026. Every feature in this plan has been screened against the four statutory criteria. Autonomous triage (OPTIMA "Epoch 2," a Class II SaMD path) is explicitly out of scope. The regulatory posture is not a constraint on the business model—it *is the business model*.

4.1 Why Non-Device CDS Is Our Primary Pathway

A 510(k) or Class II clearance is a 12–24-month, \$250K–\$2M-per-indication detour that burns runway and forces a slower, device-grade sales motion—the very dynamic that has stranded imaging-AI competitors. By contrast, software that satisfies the four statutory Non-Device criteria may be deployed into any CDS-Hooks-capable EHR today, iterated continuously, and procured at SaaS speed. For a company whose moat is speed, capital efficiency, and a federated data network that compounds with every new site, the Non-Device pathway is not merely permissible—it is strategically optimal (Figure 4.1).

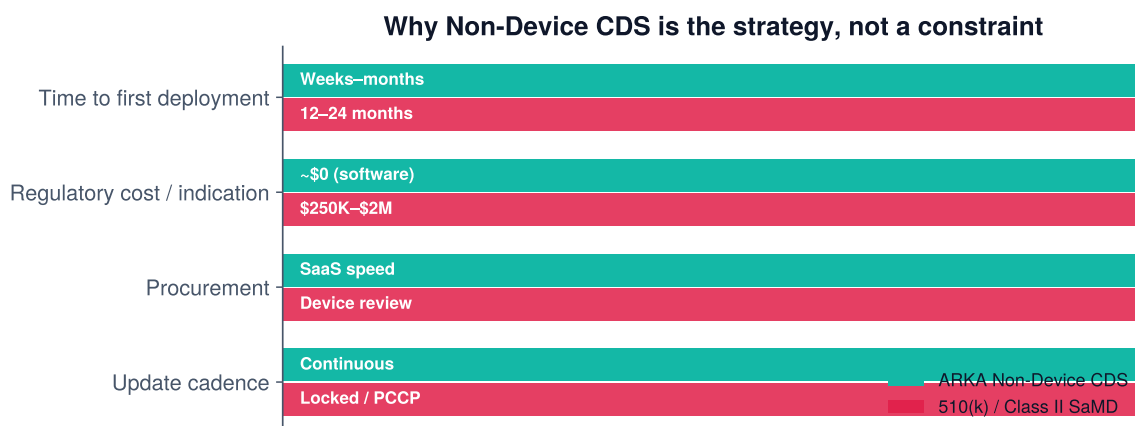


Figure 4.1. Why Non-Device CDS is the strategy, not a constraint.

Table 4.1. Regulatory pathway comparison—ARKA's choice in context.

Pathway	Typical use	Time	Cost
Non-Device CDS (ARKA)	Explainable, recommendation-only, structured-data CDS	Ship today	Software-only
510(k)	Substantially equivalent Class II device	6–12 mo	\$250K–\$1M
De Novo	Novel low/moderate-risk device, no predicate	12–18 mo	\$0.5–2M
PMA	High-risk Class III device	1–3+ yr	\$5M+

4.2 The Four Statutory Criteria and ARKA's Compliance

Section 520(o)(1)(E) excludes software from the definition of a medical device when it satisfies four criteria. Arka satisfies each one *structurally*—at the level of code, not merely of policy—and the compliance is verifiable in the repository's regulatory-rationale memo.

Table 4.2. The four §520(o)(1)(E) criteria and ARKA's code-level compliance.

Criterion	Statutory requirement	How ARKA satisfies it
1 --- Not device data	No analysis of medical images, IVD signals, or physiological waveforms	Consumes <i>only</i> structured FHIR; a CI guard blocks image/signal libraries from CDS scope; the DICOM viewer is scoped out
2 --- Basis in published evidence	Recommendations grounded in published or regulatory sources	Every card carries a structured <code>medicalBasis</code> ; a card cannot ship without it (<code>assertMedicalBasis</code>)
3 --- Recommendation, not directive	Supports a decision; does not replace clinical judgment	Non-blocking CDS Hooks; the override leads with “Clinical judgment based on findings not captured in chart”
4 --- Independent review	The clinician can independently review the basis	SHAP attributions and citations surfaced on demand; rules-only fallback when ML is offline

To make the compliance concrete, consider how each criterion is satisfied in practice. **Criterion 1 (not device data):** when a clinician orders “CT abdomen,” AII reads the structured FHIR problem list, prior-study metadata, and lab values—never the DICOM pixels; the imaging viewer is deliberately scoped out of the CDS service, and a CI guard fails any build that imports an image-processing library into CDS scope. **Criterion 2 (published evidence):** the card for that order carries a structured `medicalBasis` citing the ACR/GRADE evidence for abdominal CT in the given presentation, and the build pipeline's `assertMedicalBasis` check rejects any card lacking one. **Criterion 3 (recommendation, not directive):** the card cannot block the order—CDS Hooks has no block primitive and ARKA adds none—and the override path leads with “Clinical judgment based on findings not captured in chart.” **Criterion 4 (independent review):** the clinician can expand the SHAP attributions and citations on demand, and if the ML layer is offline the deterministic rules core renders the same cited recommendation, so the basis is always independently reviewable.

The design principle that guarantees Criterion 4. Because the deterministic v1 core can render the entire recommendation—score, factors, and citations—without the ML layer, a clinician can *always* reach and review the published basis. ML refines; it never originates. This is what makes the independent-review criterion an architectural invariant rather than a documentation promise.

4.3 The Hard NO-List, YES-List, and Boundary Lint

To keep the Non-Device boundary from eroding feature-by-feature, Arka maintains an engineering guardrail that is binding on every code review and product decision.

Table 4.3. The engineering guardrail: what ARKA will and will not do.

The hard NO-list	The hard YES-list
No pixel-level image analysis	Non-Device banner on every page
No IVD signal processing	Standard CDS card footer
No time-critical alerting acted on before clinician review	Evidence citation on every recommendation
No autonomous order placement	SHAP / per-feature explanation on demand
No pediatric-dosing or oncology-regimen SaMD categories	Scope-boundary file updated for every new in-scope directory
No patient-direct diagnostic outputs	Every OPTIMA path output carries rationale + Q -value provenance
No trajectory-simulator prediction of a time-critical event	Per-objective weight vector exposed for audit
No OPTIMA Epoch 2 autonomous triage	Affirmative clinician click required for every recommendation

A continuous-integration “boundary lint” enforces these rules automatically: no file under the OPTIMA engine directory may import an out-of-scope path, and any new card or API field is rejected if its name or schema implies pixel analysis, acute-event prediction, or autonomous submission. The guardrail is code, not a memo.

4.4 The Four-Event Regulatory Earthquake

The regulatory ground shifted four times, and every shift favors Arka (Figure 4.2).

- PAMA Appropriate Use Criteria—rescinded.** The program that would have forced clinicians to consult imaging decision support (42 CFR 414.94) was rescinded by CMS in the CY2024 rule. The regulatory stick is gone, so adoption now rides on value and culture—the revenue-and-education space Arka fills, and the reason ARKA-ED exists.
- CMS-0057-F—live.** Specific (non-boilerplate) denial reasons plus 72-hour/24-hour SLAs (January 2026), and a FHIR-native Prior Authorization API—by construction the HL7 Da Vinci PAS implementation guide—by January 2027. Arka ships it today; this is the forcing function for the payer-first motion.
- FDA’s January 6, 2026 Non-Device CDS final guidance—broadening.** It clarified the §520(o)(1)(E) boundary Arka was built inside, confirming the no-510(k) path for explainable, recommendation-framed, structured-data CDS.
- AHIP pledge + state gold-card laws.** AHIP’s 2025 commitment to 80% real-time electronic prior-authorization responses, plus proliferating state gold-card statutes, make Arka’s forward-looking gold-card scorer and specific cited reasons a compliance necessity.

The four-event regulatory earthquake — every shift favors ARKA

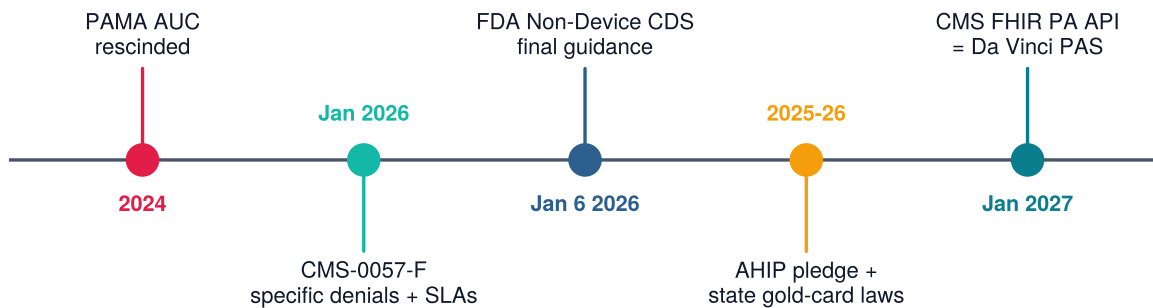


Figure 4.2. The four-event regulatory earthquake.

4.5 HIPAA and Privacy: Federated Learning, No Raw PHI Moves

Arka's engine improves across institutions via **federated learning with differential privacy**—FedAvg/FedProx with secure aggregation and a per-institution ϵ -budget ledger. Only differentially private gradient aggregates cross the network; a patient record never does. Logs are redacted to hashed identifiers and age buckets. The result is a compounding accuracy advantage—a labeled, payer-adjudicated, multi-institution outcome dataset that money cannot buy and a foundation model cannot synthesize—achieved without ever creating the HIPAA exposure that a centralized data lake would.

Table 4.4. Privacy-by-architecture primitives.

Primitive	What it guarantees
Federated learning (FedAvg/FedProx)	Models train where the data lives; no row-level PHI leaves the institution
Differential privacy (ϵ-budget ledger)	Mathematical bound on what any released aggregate can reveal ($\epsilon = 1.0$, $\delta = 10^{-6}$ /week)
Secure aggregation	The coordinator never sees an individual institution's gradient in the clear
Log redaction	Identifiers reduced to salted hashes and age buckets before any logging

4.6 Easy Integration Over Complex Tools

A proprietary integration is a multi-quarter security review that most startups never clear. Arka ships on the rails Epic and Cerner already run, so it presents to hospital IT as a *standard application*, not a bespoke risk.

4.7 CMS-0057-F: A Forcing Function in ARKA's Favor

Table 4.5. Standards-native integration surface.

Standard	Role in ARKA
HL7 CDS Hooks 2.0	Fires ARKA at order-select, encounter-start, and other decision points; non-blocking by spec
SMART-on-FHIR (R4)	Authenticated, scoped access to the structured patient context ARKA reasons over
Da Vinci CRD	Coverage Requirements Discovery—surfaces the payer’s rules at the point of order
Da Vinci DTR	Documentation Templates and Rules—the pre-filled FHIR Questionnaire that completes the packet
Da Vinci PAS	Prior Authorization Support—the real-time decision with specific, cited reasons; the CMS-0057-F endpoint

CMS-0057-F requires, by January 1, 2026, specific denial reasons and 72-hour/24-hour decision SLAs, and by January 1, 2027, a FHIR-native Prior Authorization API—by construction the Da Vinci PAS implementation guide. A payer that adopts Arka receives a drop-in compliance package that *also* produces ROI; a payer that builds in-house faces an eight-figure, multi-year FHIR project. CMS’s own estimate places ten-year industry savings from the rule at roughly \$15B. Arka converts a compliance obligation into a revenue engine—the cheapest path to the mandate and the only one that pays for itself.

A parallel signal reinforces the direction of travel: CMS’s **WISeR Model**, launching across six states, explicitly applies technology-enabled prior authorization and pre-payment review to services that historically did not require it—image-guided spine procedures, epidural steroid injections, and similar. WISeR is the regulator telling commercial plans that pre-payment intelligence will become the default. Plans without a real-time, evidence-driven decision layer on imaging will spend the next decade either over-paying or fighting clawbacks; Arka is that layer.

4.8 State Gold-Card and Appeal Law

CMS-0057-F’s specific-denial regime, combined with proliferating state gold-card laws, makes *defensible, cited denial reasons* a legal necessity rather than a convenience. Arka produces the specific reason code as a byproduct of scoring the order—the very artifact that prevents the denial also satisfies the statute. The forward-looking gold-card scorer computes a provider’s approval rate for each CPT × payer combination (Wilson lower-bound, $n \geq 20$) and auto-approves eligible orders before a prior authorization is ever created, mirroring the industry-standard $\geq 92\%$ -over-two-years gold-card logic but surfaced *forward* instead of mailed retroactively. The net effect on the order funnel is that 35–40% of orders clear automatically and never enter a review queue—which is precisely where the administrative savings live (Figure 4.3).

The regulatory thesis in one sentence. Arka is the only imaging-appropriateness platform that is simultaneously (a) free of the 510(k) detour, (b) HIPAA-safe by architecture, (c) the drop-in answer to the CMS-0057-F mandate, and (d) the producer of the cited denial reasons that state law now requires—and it achieves all four without analyzing a pixel or predicting a time-critical event.

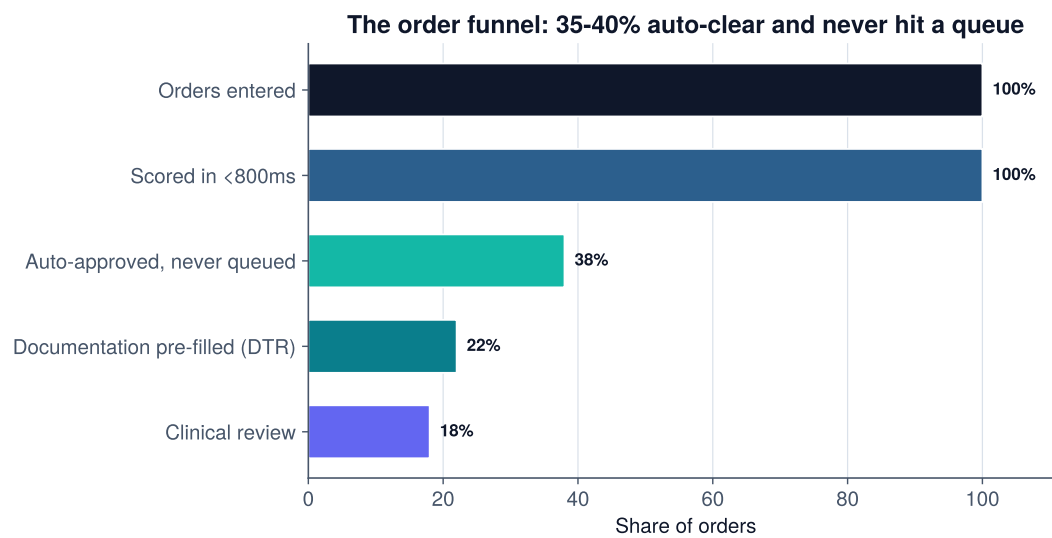


Figure 4.3. The order funnel—35-40% of orders auto-clear and never hit a review queue.

5 CHAPTER 5

Return on Investment and the Path to Phase Up

This chapter is the financial heart of the plan. It presents detailed models demonstrating Arka's exceptional return on investment across all three business lines—clinical decision support (ARKA-CLIN), medical education (ARKA-ED), and payer intelligence (ARKA-INS)—and then lays out, explicitly and in order, *how Arka plans to phase up*: from product, to engine, to go-to-market, to revenue, to valuation. Each model is built on conservative assumptions and validated against published market data, and every ROI calculation is powered by the AIIE scoring engine.

Two complementary views of ARKA's value. We present the hospital ROI two ways. The *value-based* model (Section 5.1) quantifies the full averted-imaging value that drives the headline return. The *denial-recovery floor* (Section 5.1.4) is a deliberately conservative, line-by-line model anchored entirely to published third-party figures—the number a CFO can underwrite on day one before a single outcome study. Both point the same direction; the floor simply establishes how little has to go right for Arka to pay for itself many times over.

5.1 ARKA-CLIN: Clinical Decision Support ROI

5.1.1 The Problem, Quantified

Medical imaging has become the invisible epidemic of modern healthcare. The United States performs approximately 400 million diagnostic imaging studies annually, generating roughly \$150 billion in direct costs—yet the literature is unambiguous that approximately 30–35% of all imaging is clinically inappropriate or unnecessary.

Table 5.1. The national imaging-waste opportunity.

Metric	Value
Total US imaging studies / year	400,000,000
Inappropriate imaging rate	35%
Inappropriate studies	140,000,000
Average study cost	\$800
Annual waste from inappropriate studies	\$112,000,000,000
Additional waste from redundant studies (20%)	\$24,000,000,000
Total annual imaging waste	\$136,000,000,000

This is not merely a financial problem. Inappropriate imaging increases radiation exposure, delays appropriate care, increases patient anxiety, and consumes clinical time. CT imaging, despite being only ~12% of studies nationally,

accounts for a disproportionate share of waste because average costs exceed \$3,000 per study—which is exactly why Arka’s appropriateness scoring concentrates its highest-impact attention on advanced modalities (Figure 5.1).

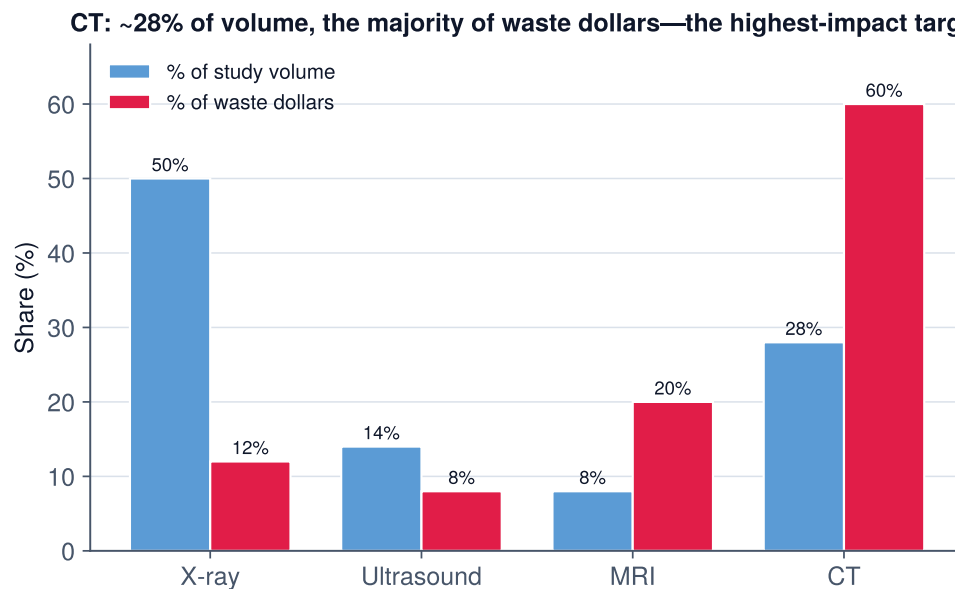


Figure 5.1. CT and MRI concentrate the waste dollars—the highest-impact target for AIIE.

5.1.2 The Six Payer Bleed Points

On the payer side, the same imaging line leaks money at six distinct points, each of which Arka addresses at the moment of order entry: (1) prior-authorization administrative overhead (~\$13B/yr); (2) appeal-overtake losses (80.7% of appealed Medicare-Advantage denials are overturned); (3) site-of-service leakage (~\$14M/yr per million lives); (4) inappropriate and low-value imaging paid as claims (30–35%); (5) the incidental-findings cascade (\$460–\$2,155 per finding); and (6) fraud, waste, and abuse on imaging that does not require prior authorization today. These pools are quantified in Section 5.3.

5.1.3 Detailed ROI Calculation: A Tier-2 Hospital (325 Beds)

We perform a complete, step-by-step ROI calculation for a representative Tier-2 hospital: a regional medical center with 325 beds and approximately 100,000 annual imaging studies. Every calculation is shown explicitly to ensure transparency and reproducibility.

Step 1 — Baseline volume and preventable-study identification.

$$\begin{aligned} \text{Inappropriate studies} &= 100,000 \times 0.35 = 35,000 \\ \text{Redundant studies (within 90 days)} &= 100,000 \times 0.20 = 20,000 \\ \text{Total preventable studies} &= 35,000 + 20,000 = 55,000 \end{aligned}$$

Step 2 — Cost structure by modality. Establishing the cost base using Medicare average allowed amounts and hospital billing practices:

Step 3 — ARKA impact parameters. Arka’s net capture rate reflects algorithm sensitivity (~85%) × workflow integration (~85%) × post-override retention (~83%) ≈ 60%. Inappropriate imaging is reduced by 45% (central case), and redundancy capture runs at 65%.

Table 5.2. Tier-2 hospital imaging cost structure (100,000 studies).

Modality	% volume	Studies/yr	\$/study	Total cost
X-ray (chest, extremities)	50%	50,000	\$285	\$14,250,000
Ultrasound (general, vascular)	14%	14,000	\$400	\$5,600,000
CT (brain, chest, abdomen)	28%	28,000	\$3,275	\$91,700,000
MRI (brain, MSK, cardiac)	8%	8,000	\$1,325	\$10,600,000
Total	100%	100,000		\$122,150,000

Table 5.3. Studies prevented (central case).

Component	Calculation	Result
Inappropriate (baseline)	$100,000 \times 0.35$	35,000
After 45% reduction	$35,000 \times 0.45$	15,750
After 60% capture	$15,750 \times 0.60$	9,450
Redundant (baseline)	$100,000 \times 0.20$	20,000
After 65% capture	$20,000 \times 0.65$	13,000
Total studies prevented	$9,450 + 13,000$	22,450

Step 4 — Cost-savings calculation. Weighting prevented studies by modality yields \$27,423,875 of gross averted cost. We conservatively reduce this to **\$23,800,000** to account for incomplete payer capture, contract-rate variation, and patient out-of-pocket reduction.

Step 5 — ARKA-CLIN fee. Arka employs a risk-sharing, value-based model with a base fee of \$0.375/study (= \$37,500/yr), or roughly \$375,000 over a three-year agreement with volume discounting. The per-study rate steps down by annual volume tier so that larger systems capture scale economics while Arka retains share of the value created (Table 5.4).

Table 5.4. ARKA-CLIN volume-based pricing tiers.

Tier	Annual studies	\$/study	Indicative annual fee
Critical-access	<10,000	\$0.45	<\$4,500
Community	10K–75K	\$0.40	\$4,000–\$30,000
Regional	75K–200K	\$0.375	\$28,000–\$75,000
Large / IDN	200K–400K	\$0.34	\$68,000–\$136,000
Academic / network	>400K	\$0.30 (+ enterprise)	\$120,000+

Step 6 — Net savings and ROI.

Net savings (3-year) = \$23,800,000 – \$375,000 = \$23,425,000

ROI = \$23,425,000/\$375,000 = **6,247%**

Payback period = \$375,000/\$23,800,000 ≈ **17 days**

Worst case still wins. If Arka achieves only 60% of projected savings (lower compliance, payer friction, partial implementation), realized savings are \$14,280,000, net savings \$13,905,000, and ROI still exceeds

450%. Even the pessimistic corner returns more than \$4.50 of benefit for every dollar invested over three years.

5.1.4 The Denial-Recovery Floor: A CFO Can Sign This on Day One

The value-based model above is compelling but depends on a utilization-reduction assumption. The *denial-recovery floor* makes no such assumption: it counts only revenue the hospital already earned but failed to collect because the documentation a payer required was missing. Modeled for a regional group running 120,000 advanced imaging studies per year, with every figure drawn from published third-party sources (Figure 5.2):

Table 5.5. Denial-recovery floor (regional group, 120,000 advanced studies/yr).

Lever	Mechanism (conservative)	Annual impact
A. Denial recovery	$120,000 \times 25\% \text{ PA-required} \times 25\% \text{ denied} \times 86\% \text{ avoidable} \times 60\% \text{ captured} \times \900	$\approx \$3.48\text{M}$
B. Throughput defense	+2% completed advanced studies on a high-margin line	$\approx \$0.54\text{M}$
C. Rework labor avoided	$\sim 3,870 \text{ denials avoided} \times \$50 \text{ blended rework cost}$	$\approx \$0.19\text{M}$
D. Admin headcount redirected	2–3 prior-auth FTEs redirected $\times \$75\text{K loaded}$	$\approx \$0.20\text{M}$
Gross annual benefit		$\approx \$4.41\text{M}$
Less ARKA fee		$-\$0.40\text{--}0.70\text{M}$
Net Year-1 benefit		$\approx \$3.7\text{--}4.0\text{M}$

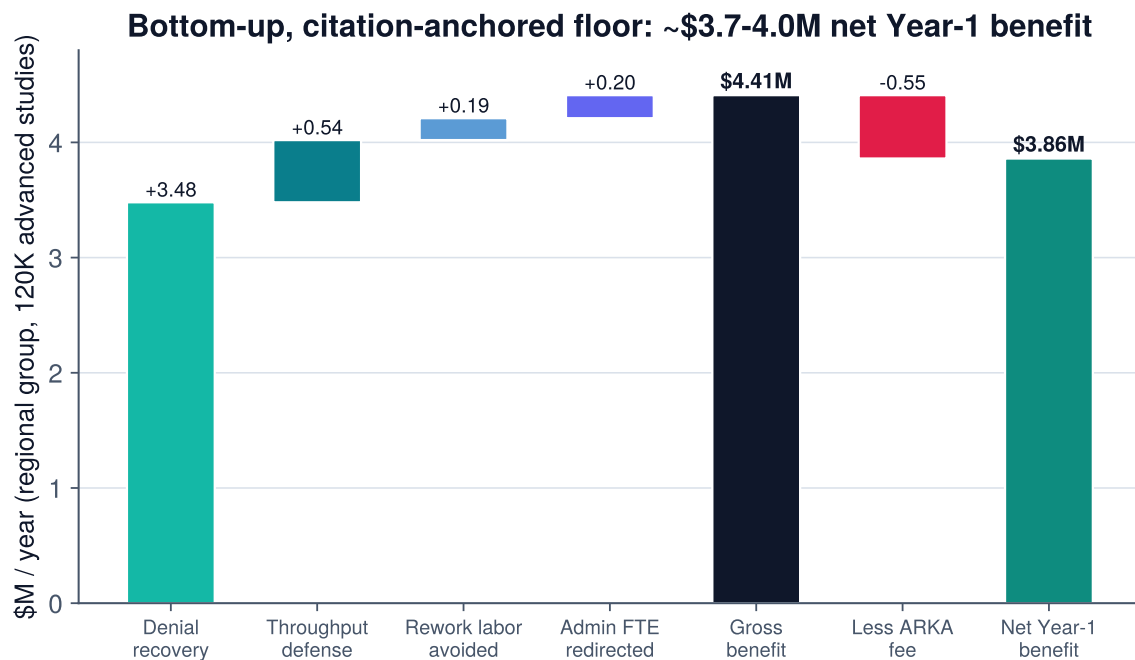


Figure 5.2. The bottom-up, citation-anchored denial-recovery floor.

5.1.5 Five-Tier Hospital ROI Model

Arka scales across the hospital spectrum, from 15-bed critical-access hospitals to 750-bed academic medical centers. The consistency of ROI across tiers—approximately 190% per year, or ~6,300% over three years—demonstrates that the value proposition is universal (Table 5.6, Figure 5.3).

Table 5.6. ROI across all five hospital tiers.

Tier	Beds	Studies/yr	Prevented	Savings/yr	3-yr ROI
Tier 1: Academic	750	350,000	78,750	\$83.2M	189,947%
Tier 2: Regional	325	100,000	22,450	\$23.8M	190,300%
Tier 3: Community	150	45,000	10,103	\$10.7M	190,049%
Tier 4: Small	50	15,000	3,368	\$3.57M	190,300%
Tier 5: Critical-access	15	4,500	1,010	\$1.07M	190,300%

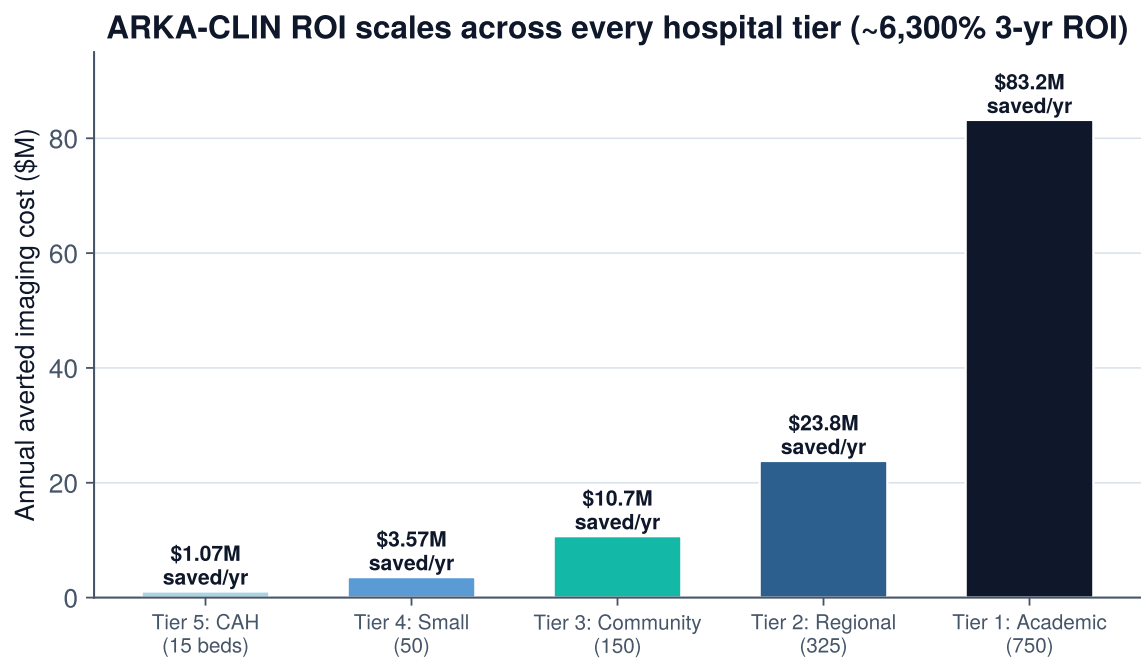


Figure 5.3. Annual averted imaging cost scales across every hospital tier.

5.1.6 Sensitivity Analysis: Robustness to Parameter Variation

The ROI calculation depends on three parameters: the inappropriate-reduction rate, the capture rate, and cost per study. Across all reasonable combinations, the value proposition remains extraordinarily robust: even the worst-case corner (20% capture, 20% reduction) exceeds 450% three-year ROI (Figure 5.4).

5.1.7 Market-Capture Opportunity

There are approximately 6,150 hospitals in the United States: ~325 major teaching hospitals (Tier 1), ~1,250 large community hospitals (Tiers 2–3), ~3,575 small/critical-access hospitals (Tiers 4–5), and ~1,000 specialty facilities. Conservative penetration: 5% in Years 1–2 (~308 hospitals), 15% in Years 3–4 (~923), and 30% by Year 5+ (~1,845).

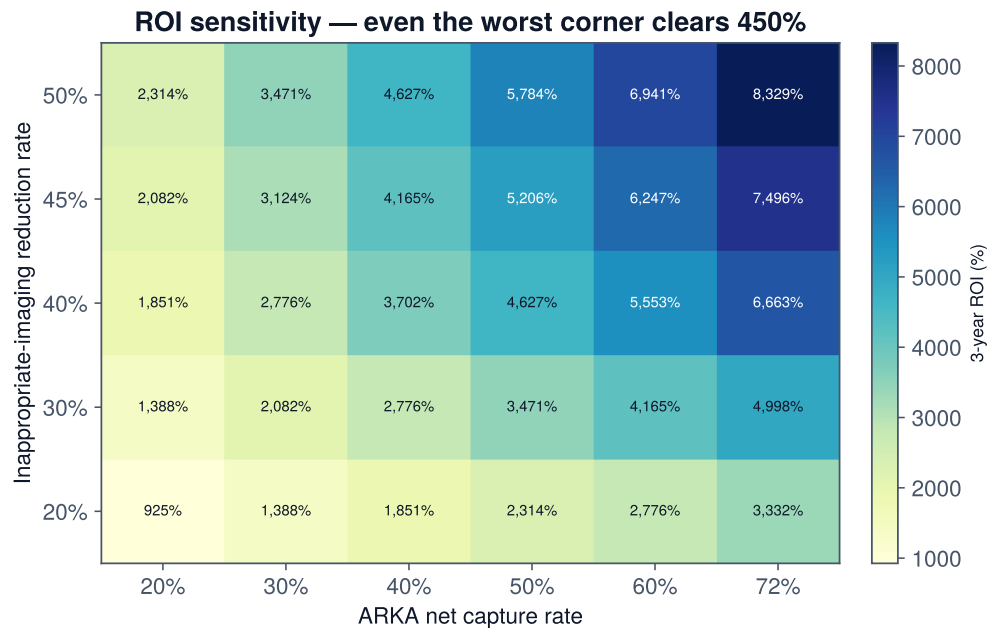


Figure 5.4. ROI sensitivity heat map—even the worst corner clears 450%.

5.1.8 Rural and Resource-Aware Market Expansion

A distinctive extension of ARKA-CLIN is its resource-aware mode (delivered as a deployment configuration rather than a separate product), which tailors recommendations to the capability of low-resource and rural sites. Where a 25-bed critical-access hospital lacks an on-site MRI or after-hours radiology coverage, AIIE incorporates local resource availability into the Pareto optimization—recommending the modality that is both appropriate and *actually obtainable*, and flagging when transfer to a higher-capability center is the genuinely appropriate path. This expands ARKA's addressable market into the ~1,800 rural and critical-access hospitals that incumbents ignore, converts a clinical-equity story into a commercial one, and strengthens the federated data moat with presentations underrepresented in urban academic datasets. Rural deployment is also a powerful regulatory and public-relations asset: it demonstrates that Arka's value is access and appropriateness, not denial.

5.2 ARKA-ED: Medical Education ROI

The US medical-education market is a multi-billion-dollar annual opportunity spanning students, residents, CME credits, and hospital educational licenses—and ARKA-ED is powered by the very same AIIE engine, so its content improves automatically as ARKA-CLIN data accrues.

Table 5.7. ARKA-ED addressable market.

Segment	Users	Avg spend	Market size
Medical students (4-yr)	95,000	\$450	\$427.5M
Residents (5-yr)	155,000	\$2,200	\$341.0M
CME credits (physicians)	1,100,000	\$2,800	\$3,080.0M
Hospital educational licenses	6,150	\$75,000	\$461.3M
Total addressable market			\$4,309.8M

ARKA-ED revenue builds from B2C subscriptions (student and resident tiers) and B2B institutional licenses (medical schools, hospital education), reaching \$83.3M by Year 5 (Figure 5.5).

Table 5.8. ARKA-ED revenue by stream, Years 2–5.

Stream	Year 2	Year 3	Year 4	Year 5
Student Basic subscriptions	\$2.78M	\$6.26M	\$11.14M	\$16.70M
Student Pro subscriptions	\$1.76M	\$4.41M	\$8.23M	\$12.35M
Resident programs	\$2.25M	\$5.25M	\$9.00M	\$14.25M
Small med-school licenses	\$1.25M	\$2.25M	\$3.75M	\$5.50M
Large med-school licenses	\$2.25M	\$4.50M	\$7.50M	\$12.00M
Hospital education licenses	\$3.75M	\$9.38M	\$15.00M	\$22.50M
Total ARKA-ED	\$14.05M	\$32.05M	\$54.62M	\$83.30M

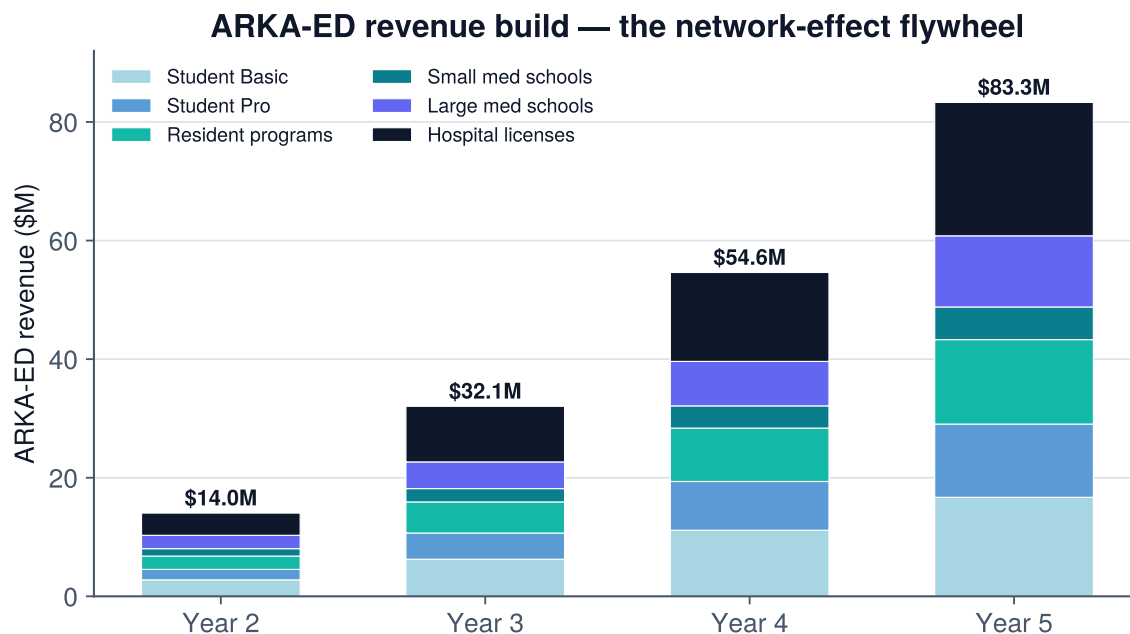


Figure 5.5. ARKA-ED revenue build—the network-effect flywheel.

The learning-loop flywheel. Clinical data from ARKA-CLIN drives educational content, which trains a new generation of clinicians who make better decisions, generating richer data—a virtuous cycle that compounds with each cohort and is impossible for a standalone education vendor to replicate. Graduating residents then request ARKA-CLIN at their future employers, closing the adoption loop.

5.3 ARKA-INS: Payer Intelligence ROI

ARKA-INS monetizes the aggregated, de-identified clinical intelligence from ARKA-CLIN across four revenue streams—payer benchmarking, pharma intelligence, clinical-trials intelligence, and prior-authorization analytics—

reaching \$17.1M by Year 5 (Figure 5.6). Crucially, ARKA-INS carries zero revenue in Year 1 by design: 12–18 months of ARKA-CLIN deployment must accrue before the data and relationships have value.

Table 5.9. ARKA-INS revenue by stream, Years 1–5.

Stream	Y1	Y2	Y3	Y4	Y5
Payer benchmarking	\$0	\$0.9M	\$1.4M	\$1.85M	\$2.3M
Pharma intelligence	\$0	\$1.4M	\$2.1M	\$3.2M	\$4.8M
Clinical-trials intelligence	\$0	\$0.64M	\$1.2M	\$2.0M	\$3.0M
PA analytics	\$0	\$0	\$2.2M	\$4.15M	\$7.0M
Total ARKA-INS	\$0	\$2.94M	\$6.9M	\$11.2M	\$17.1M

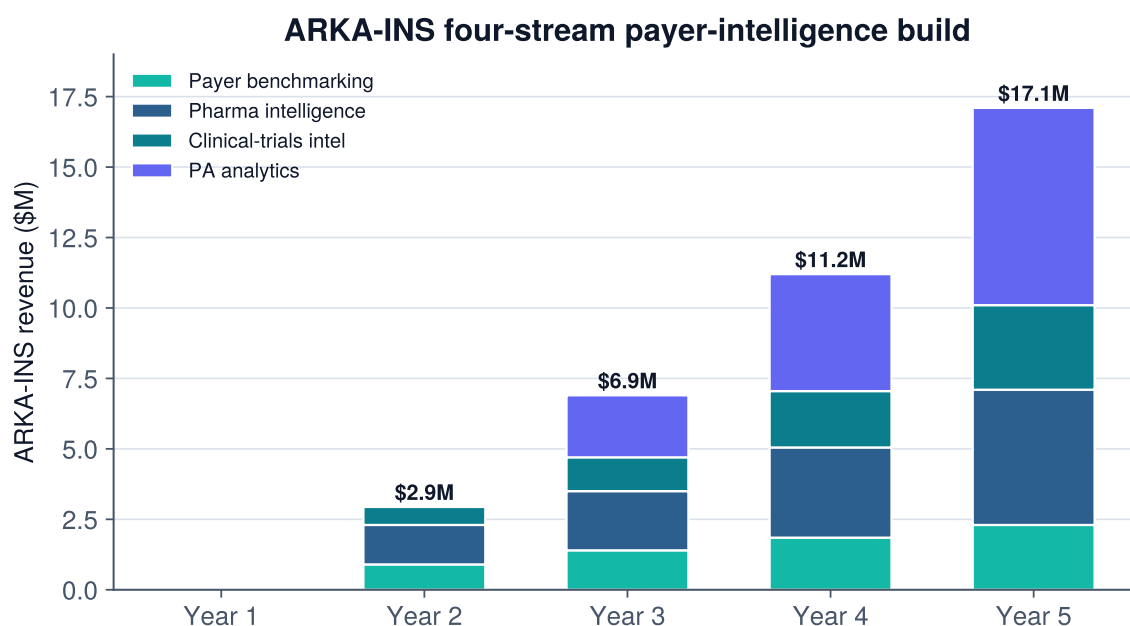


Figure 5.6. ARKA-INS four-stream payer-intelligence build.

5.3.1 The Four Revenue Streams in Detail

Stream 1 — Payer benchmarking analytics. Health plans need to understand their networks' imaging-appropriateness performance relative to competitors and national benchmarks. ARKA provides regional benchmarks, facility- and provider-level inappropriate-imaging rates, and trend analysis. Pricing: \$25,000–\$100,000/yr per payer by network size.

Stream 2 — Pharma intelligence. Companies launching diagnostic drugs or devices need market-penetration and adoption insight for competitive imaging modalities and biomarkers. Pricing: \$300,000–\$500,000 per six-month engagement.

Stream 3 — Clinical-trials intelligence. Trial sponsors need to identify patient populations meeting imaging criteria and estimate recruitment feasibility across regions. Pricing: \$50,000 setup plus \$10–\$25 per patient screened.

Stream 4 — Prior-authorization analytics. Plans increasingly need data-driven PA policies: evidence-based criteria updated quarterly, audit tools, impact modeling, and appeals analysis. Pricing: \$200,000–\$500,000/yr per large payer.

The conservative build per stream reflects the time required to accrue data and relationships. Payer benchmarking ramps from ~15 payers (avg \$60K) for \$0.9M in Years 2–3 to ~40 payers for \$2.3M by Year 5. Pharma intelligence grows from 4 engagements ($\times \$350K = \$1.4M$) to 12 ($\times \$400K = \$4.8M$). Clinical-trials intelligence scales from 8 trials (\$640K) to 25 trials (\$3.0M). PA analytics, which depends most on accrued adjudication data, begins in Year 3 at 8 payers ($\times \$275K = \$2.2M$) and reaches 20 payers ($\times \$350K = \$7.0M$) by Year 5. Each stream monetizes the same aggregated, de-identified intelligence that ARKA-CLIN generates—without ever exposing an individual patient record—which is why ARKA-INS is pure margin layered on an asset the provider business already produced.

5.3.2 A Worked Payer Example

A 1-million-life regional commercial plan. At 15–25 imaging PAs per 1,000 members, the plan files 15,000–25,000 imaging PAs per year. ARKA auto-approves 38% (removing \$2–4M of G&A), prevents the wrong denials that drive an 80.7%-overturn appeals tax, routes 25% of HOPD-bound studies to a cheaper in-network site (the \$14M site-of-service line), and drops inappropriate studies straight to medical-loss ratio. At the per-1,000-lives economics below, that is roughly **\$10.9M of net annual value** against a \$0.40 PMPM (\$4.8M) fee—before counting STAR-rating improvement worth \$80–120M per point.

Beneath the company-level revenue sits the per-member economics that a payer underwrites. Modeled conservatively per 1,000 covered lives, ARKA-INS returns roughly \$15,667 of gross annual value against a \$4,800 fee (at \$0.40 PMPM)—a net of ~\$10,867, or about $2.3\times$ year-one ROI before soft savings:

Table 5.10. ARKA-INS value per 1,000 covered lives (conservative).

Line	Mechanism	Per 1,000 lives
PA admin saved	38% auto-approval removes human touch	\$31
Appeal volume reduction	Fewer wrong denials to appeal	\$15
Appeal-overturn loss reduction	Cited reasons survive challenge	\$28
Inappropriate-imaging avoidance	Pre-order appropriateness drops to MLR	\$1,512
Site-of-service leakage capture	Cheaper in-network site at order	\$14,000
Cascade-cost avoidance	Fewer incidental-finding cascades	\$76
FWA / improper-payment reduction	Pre-payment intelligence	\$5
Gross per 1,000 lives		≈ \$15,667
Less ARKA fee (\$0.40 PMPM)		–\$4,800
Net per 1,000 lives		≈ \$10,867

Scaled conservatively, that is roughly **\$543M of net Year-1 value for a 50-million-life national carrier** (\$815M in the aggressive case), before accounting for STAR-rating improvement worth \$80–120M per STAR point on a 500,000-life plan. The payer ROI is not a rounding error—it is a board-level number (Figure 5.7).

5.4 How ARKA Plans to Phase Up

Arka's growth is not a single bet but a deliberately sequenced set of phases that compound. There are four phasing axes—product, engine, go-to-market, and revenue—and they are timed to reinforce one another (Figure 5.8).

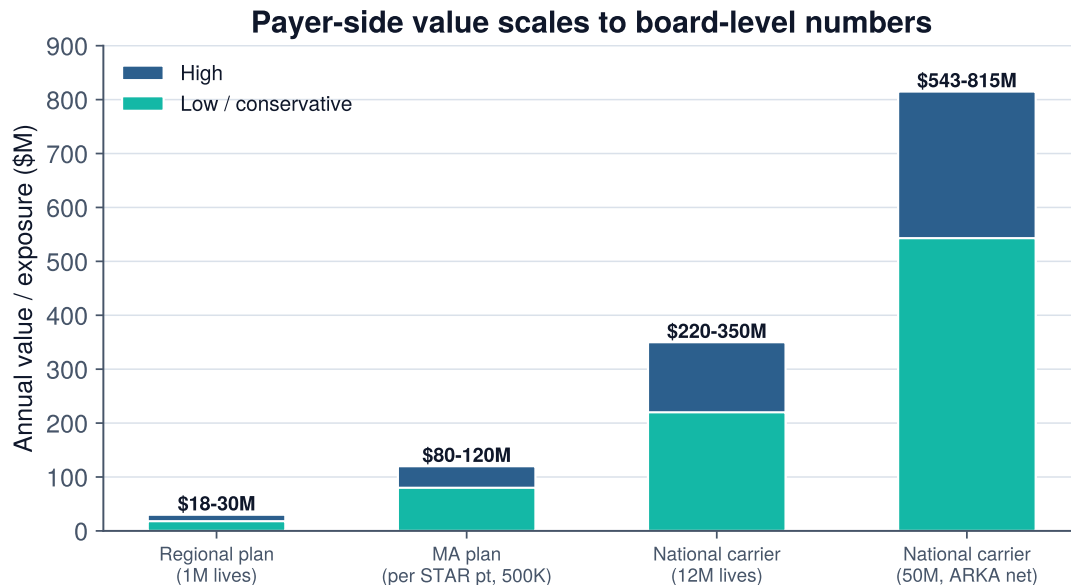


Figure 5.7. Payer-side value scales from regional plans to board-level national-carrier numbers.



Figure 5.8. The phase-up ladder—product waves, engine epochs, and the revenue ramp, timed to reinforce one another.

5.4.1 Phase-Up Axis 1 --- Product: The Three Waves

ARKA-CLIN leads (highest TAM, fastest path to revenue, the data source for everything else). ARKA-ED follows once clinical data exists to power realistic case simulation, creating the network-effect flywheel and a culture of appropriateness that no longer has a regulatory stick behind it. ARKA-INS comes last because it requires 12–18 months of accrued ARKA-CLIN data and rides the January-2027 CMS-0057-F forcing function.

5.4.2 Phase-Up Axis 2 --- Engine: AIIE v1 to AIIE-OPTIMA (O-0 → O-5)

The engine upgrades behind a feature flag, with each gate proven before advancing and every phase remaining Non-Device.

Table 5.11. The AIIE-OPTIMA execution sequence.

Phase	Deliverable	Gate before advancing
O-0	Dispatcher scaffold behind the ARKA_ENGINE flag	v1 output byte-identical; fallback proven
O-1	Belief + Value-of-Information (D19) and HGNN encoder	Calibration within target on the retrospective lake
O-2	Simulator + MCTS + bandit (D20), with constraints	Negative test: no acute-event semantics anywhere
O-3	Federated network (D21) + DR-OPE (D22)	ϵ -ledger enforced; DR estimator validated
O-4	Shadow mode at three sites (D24)	30-day shadow; calibration drift <5%
O-5	Mirror → opt-in A/B → default-with-override	Pre-registered $\geq 15\%$ dose / $\geq 20\%$ cost at non-inferior yield
O-6 (excluded)	Epoch 2 autonomous triage	Would require a 510(k)---not pursued

5.4.3 Phase-Up Axis 3 --- Go-to-Market: Payer-First and the 90-Day Pilot

The wedge is the regional or mid-sized group most desperate for new revenue, plus a payer-first motion where CMS-0057-F is closing the deal. Each engagement begins with a self-funding 90-day pilot:

Table 5.12. The 90-day pilot.

Window	Activity
Days 0--14 --- Discovery	12 months of de-identified claims/PA data; agree four success metrics
Days 15--30 --- Sandbox	CDS Hooks discovery endpoint scoped to the network; one specialty opts in; no PHI moves
Days 31--60 --- Live pilot	Live scoring on real orders; ROI dashboard logs validation events; weekly business review
Days 61--90 --- Expand or no-go	Line-item ROI report; go/no-go; if go, a 12-month rollout ahead of the mandate

5.4.4 Phase-Up Axis 4 --- Pricing: Three Accretive Models

All three pricing models are gross-margin accretive to the buyer, and the company graduates customers between them as the relationship deepens.

Table 5.13. Three pricing models, all accretive to the buyer.

Model	Terms
PMPM SaaS (plans)	\$0.30–\$0.50 PMPM, 3-yr term, volume tiers; full CRD/DTR/PAS + AIIE + dashboards. At \$0.40 PMPM vs. $\sim \$15.67/\text{member}/\text{yr}$ gross value → $\sim 2.3\times$ year-1 ROI
Per-PA processed (RBMs)	\$1.50–\$3.00 per PA; Gold-Card auto-approval credited at \$0.50; lower above 5M lives
Risk share (innovative payers)	15–25% of net imaging savings, capped at \$1.00 PMPM, measured against a jointly-set baseline

5.4.5 Phase-Up Axis 5 --- Revenue: The Year 1→5 Build

The four axes compound into the consolidated ramp: \$2.1M → \$8.4M → \$28.7M → \$78.2M → \$172.4M, a 47% CAGR, with the revenue mix shifting from CLIN-dominated in Year 1 to a diversified three-pillar base by Year 5.

5.5 Multi-Phase Diversification: Why ARKA Is Resilient

Arka deliberately stages entry across three business lines to create resilience against execution risk. Even under adverse scenarios—CLIN underperformance, an ED-market failure, or a regulatory shock—the diversified base holds, with the worst case still delivering \$64.9M of Year-5 revenue (Figure 5.9).

Table 5.14. Scenario analysis: Year-5 revenue under stress.

Scenario	Assumption	CLIN	ED	INS	Total
Base case	All lines at plan	\$90.0M	\$83.3M	\$17.1M	\$190.4M
CLIN underperforms	CLIN 30%, ED 60%, INS 80%	\$21.6M	\$50.0M	\$13.7M	\$85.3M
ED market fails	ED 40%, others at plan	\$72.0M	\$33.3M	\$17.1M	\$122.4M
Regulatory shock	CLIN 25%, others at plan	\$26.0M	\$83.3M	\$17.1M	\$126.4M
Worst case	CLIN 30%, ED 40%, INS 60%	\$21.6M	\$33.3M	\$10.0M	\$64.9M

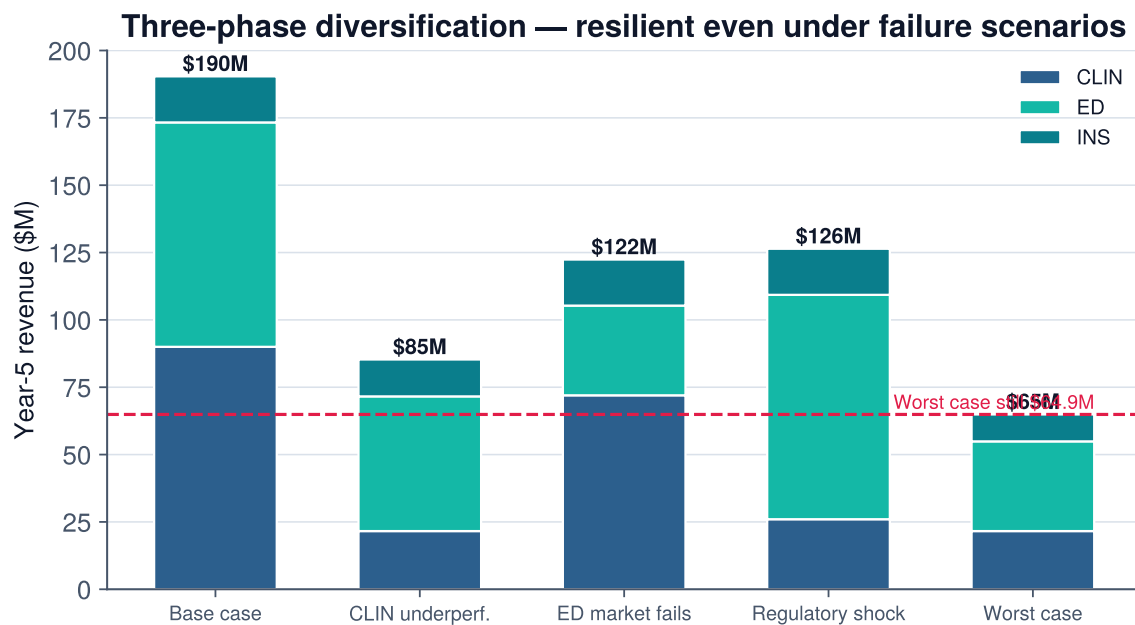


Figure 5.9. Three-phase diversification—resilient even under failure scenarios.

5.6 Consolidated Five-Year Financial Summary

Bringing the three lines together yields the consolidated pro-forma below. Arka follows the classic SaaS profitability curve—heavy early investment in product and go-to-market, gross margin expanding toward the low-80s, and oper-

ating leverage driving EBITDA margin to roughly 30% by Year 5 as revenue scales against a largely fixed engineering base.

Table 5.15. Consolidated pro-forma summary, Years 1–5 (\$M unless noted).

Line	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	\$2.1	\$8.4	\$28.7	\$78.2	\$172.4
Gross profit	\$1.6	\$6.6	\$23.0	\$63.3	\$141.4
Gross margin	78%	79%	80%	81%	82%
Sales & marketing (% rev)	95%	70%	48%	36%	30%
Research & development (% rev)	110%	55%	30%	20%	14%
General & administrative (% rev)	40%	24%	14%	10%	8%
EBITDA margin	–130%	–45%	–8%	16%	30%
Headcount (FTE)	25	55	110	190	300

The model crosses into positive EBITDA in Year 4 and exits Year 5 at a ~30% margin on \$172.4M of revenue—the combination of high growth and emerging profitability that produces a Rule-of-40 score of 112 and supports the exit multiples in Section 1.7.

5.7 The Path to Unicorn Valuation

Synthesizing the three lines, Arka reaches \$172.4M of Year-5 revenue at an 82% gross margin and a Rule-of-40 score of 112. At the SaaS norm of 8–12 $\times$ revenue—and the 15–20 $\times$ that regulatory-compliance healthcare software commands—this supports a \$2.6B–\$3.4B exit valuation and a 100–140 $\times$ return on the Series A. The ROI story is therefore fractal: it holds at the level of a single imaging order (a cited, defensible recommendation), at the level of a single hospital (6,247% three-year ROI, or a conservative \$3.7–4.0M Year-1 floor), at the level of a payer (\$543M net for a national carrier), and at the level of the company (a unicorn outcome). *Each phase funds and de-risks the next.*

The phase-up thesis. Arka does not ask investors to believe a single number. It asks them to believe a *sequence*: ship the Non-Device bilateral engine into the regional wedge, prove the denial-recovery floor on a self-funding pilot, layer ARKA-ED to build the culture and the data flywheel, ride CMS-0057-F into the payer market with ARKA-INS, and upgrade the engine to AIIE-OPTIMA behind an unchanged interface—each step compounding the last. That sequence is what turns a 9:1 hospital ROI into a unicorn.

Clinical Validation and Evidence Generation

The most common objection to any clinical-AI company is that its validation is thin. Arka addresses this objection structurally and then proves it empirically. Structurally, because the primary basis of every recommendation is a *published guideline*—the clinical evidence is carried by the literature, not invented by the model—the engine begins from a defensible position before a single prospective patient is enrolled. Empirically, Arka layers retrospective validation, a pre-registered prospective study, and continuous in-production proof of value on top of that foundation.

6.1 Why Validation Is Different for ARKA

A black-box model must prove that its outputs are correct from scratch, because nothing else vouches for them. Arka's rules-first, citation-first architecture inverts the burden: every card cites the guideline that justifies it, the deterministic core can render the decision without any ML, and the ML layer only *refines* a recommendation a guideline already supports. Validation therefore measures *calibration and impact*—is the engine well-calibrated against real decisions, and does it move the metrics that matter—rather than attempting to establish clinical truth de novo.

6.2 The Evidence Hierarchy

AIIE weights evidence by GRADE level, so stronger evidence dominates weaker evidence automatically, and temporal decay keeps the corpus current.

Table 6.1. The GRADE evidence hierarchy as implemented in AIIE.

Level	Definition	Weight
A --- High	Multiple consistent RCTs; meta-analyses of high-quality RCTs	Highest
B --- Moderate	RCTs with limitations, or high-quality observational studies	Substantial
C --- Low	Observational studies with limitations; case series	Moderate (decays with age)
D --- Very low	Expert consensus without empirical support; case reports	Minimal (high temporal decay)

6.3 Retrospective Validation on the Data Lake

Before any prospective deployment, AIIE is validated retrospectively against a de-identified data lake of historical orders, payer adjudications, and outcomes. The engine scores each historical order and its predictions are compared with the realized adjudication and downstream outcome. The key retrospective metrics are calibration (Brier

score within target), discrimination (the score's ability to separate approved from denied), and concordance with documented appropriate-use determinations. This is the gate that each OPTIMA phase must clear before advancing.

6.4 The Pre-Registered Prospective Study

Arka's prospective validation is pre-registered—endpoints and analysis plan fixed in advance—to avoid the post-hoc cherry-picking that undermines so many digital-health claims. The primary endpoints are the four shown in Figure 6.1: a reduction in the inappropriate-imaging rate, a reduction in the imaging-denial rate, a low clinician override rate (the trust signal), and an increase in documentation completeness (the mechanism behind clean claims).

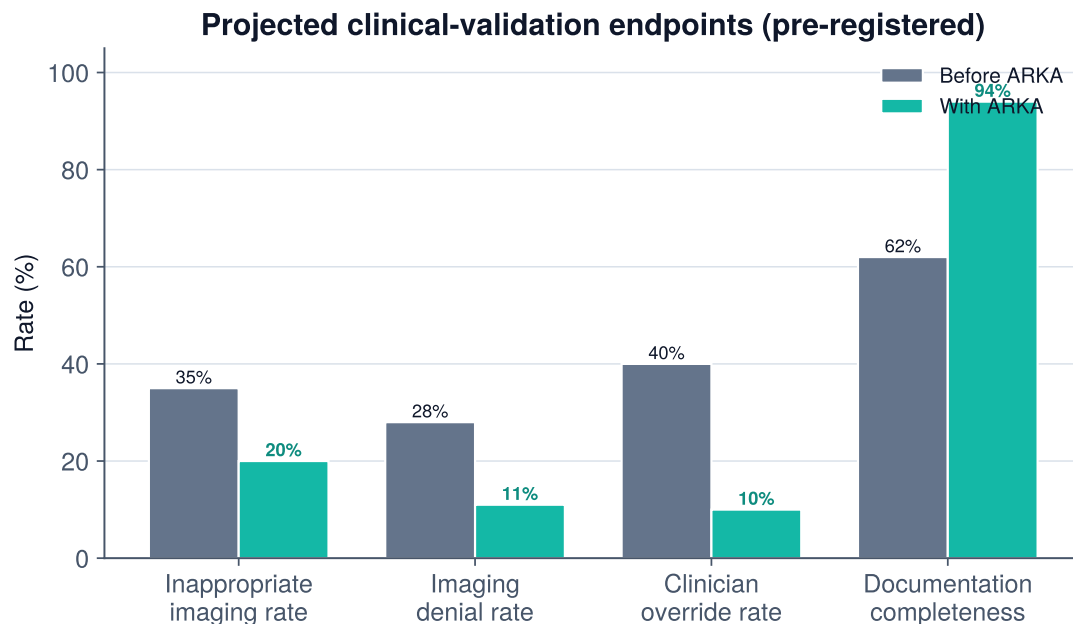


Figure 6.1. Pre-registered clinical-validation endpoints, before versus with ARKA.

For AIIE-OPTIMA specifically, the pre-registered efficacy target is the headline number that fronts every payer and ACO pitch: **≥15% cumulative radiation reduction and ≥20% cost reduction at non-inferior diagnostic yield**. Non-inferiority on yield is the safety guardrail; the radiation and cost reductions are the value.

6.5 Continuous In-Production Validation (DR-OPE)

Validation does not stop at launch. Doubly-Robust Off-Policy Evaluation re-scores the deployed policy every week against the trailing 30 days of real decisions, and any regression greater than 3% triggers an automatic rollback to the prior policy. Arka therefore does not merely *claim* it reduces cost and denials—it *measures* the effect continuously, per institution and per payer, on a live dashboard. For a chief quality officer accountable for STAR variance and an investor pricing durability, continuous statistical proof of value is the difference between a pitch and a compounding, de-risked asset.

6.6 Clinical Governance and Publication Strategy

A clinical advisory board of practicing radiologists, emergency physicians, and primary-care clinicians reviews the knowledge graph, the factor weights, and every new in-scope module before deployment; a clinical sign-off log records each approval. Arka's publication strategy targets 2–3 peer-reviewed validation studies within the first 18 months—the evidence base that converts early-adopter case studies into mainstream adoption and that anchors the Series B narrative.

The validation thesis. Arka de-risks the single most common clinical-AI objection three ways at once: the guideline carries the evidentiary burden, the retrospective lake proves calibration before launch, and DR-OPE proves impact continuously after it. Thin validation is not a risk Arka hopes to outrun—it is a risk the architecture was designed to retire.

7 CHAPTER 7

Retention Strategy: Professionals, Students, and RBMs

Adoption gets a customer in the door; retention is what compounds into a durable, expanding revenue base and a 124% net-revenue-retention figure. Arka's retention strategy is built on a single observation: clinical decision support fails not because clinicians reject *good information*, but because they reject *friction*. Arka removes friction, and then earns trust by being measurably right.

7.1 The Retention Crisis: Why Physicians Abandon CDS

The historical record of clinical decision support is a graveyard of high-adoption, low-retention deployments. Systems are mandated, used reluctantly for a quarter, overridden 90–96% of the time, and then quietly disabled. The root causes are consistent across the literature.

Table 7.1. Why clinicians abandon decision support—and how ARKA answers each.

Failure mode	Why it drives abandonment	ARKA's answer
Alert fatigue	Every deviation is flagged; the clinician deletes the app in a week	Empty cards when no guideline-anchored rule fires; a per-site bandit converges from ~15% to ~2% exploration
Workflow disruption	Pop-ups, redirects, and extra clicks slow the order	Non-blocking, in-flow, sub-800ms; zero added clicks
No clear ROI	The clinician sees cost, never benefit	A live ROI dashboard shows dollars recovered per institution
Black-box distrust	Clinicians won't follow a recommendation they can't interrogate	SHAP attributions + citations on every card
No feedback loop	Overrides vanish into the void; the system never improves	Overrides are logged and feed continuous, validated improvement

7.2 ARKA's Retention Philosophy: Earn the Interruption

The design contract is that Arka must *earn* every interruption before it makes one. A Value-of-Information floor suppresses low-yield prompts; the engine stays silent unless a published guideline fires; and when it does speak, it leads with the evidence and offers an actionable alternative rather than a scold. The override is always one neutral click and is framed as a legitimate clinical judgment, never as non-compliance. This is the direct antidote to the “COPD-patient-versus-marathoner” problem that destroys one-size-fits-all alerting.

7.3 Specific Retention Mechanisms

7.3.1 For Physicians

Arka retains physicians by making their existing workflow safer and their documentation cleaner without asking anything of them. The order that used to bounce back as a denial now clears the first time; the peer-to-peer call that used to consume an afternoon no longer happens. Arka *removes* friction rather than adding it.

7.3.2 For Students and Residents (ARKA-ED)

ARKA-ED retains learners by being the tool they trained on. A resident who learned imaging appropriateness inside ARKA-ED's realistic case simulator arrives at their first attending job fluent in the same engine that runs in the EHR—and asks for it. Retention here is generational: today's resident is tomorrow's ARKA-CLIN champion.

7.3.3 For RBMs and Utilization-Review Teams (ARKA-INS)

Arka retains payer and RBM users by inverting their economics. Reviewers report a 60% reduction in time per case and higher confidence in their decisions, because every approval is backed by the same cited math the provider saw. The per-case margin of an RBM rises the moment 35–40% of orders auto-approve, which converts ARKA-INS from a tool the reviewer tolerates into infrastructure the RBM depends on.

7.4 The Retention Flywheel

Once Arka crosses roughly 60% physician engagement—typically by Month 6—a self-reinforcing flywheel takes hold (Figure 7.1). More usage produces better data; better data produces better recommendations; better recommendations produce higher trust; higher trust produces more usage. Each turn deepens the moat and raises switching costs.

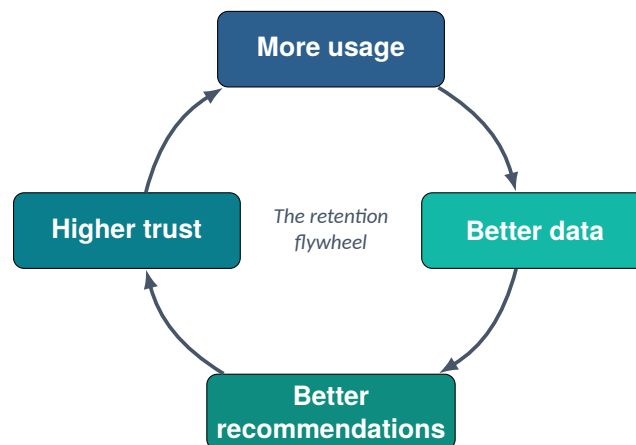


Figure 7.1. The retention flywheel—usage compounds into trust, and trust into usage.

7.5 The Customer Journey: From First Order to Renewal

Retention is best understood as a journey with predictable inflection points. In *week one*, ARKA is installed as a CDS-Hooks app and begins scoring in shadow mode—no cards surfaced, baseline metrics gathered. By *month*

one, cards go live for a single high-volume specialty; the clinical champion reviews early overrides and the bandit begins learning the site's baseline. By *month three*, auto-approval is removing 35–40% of orders from the prior-auth queue and the rev-cycle team sees the first clean-claim lift. At *month four*, the first CFO ROI review puts recovered dollars on the table—the moment the relationship shifts from “pilot” to “infrastructure.” By *month six*, physician engagement crosses 60%, the flywheel engages, and expansion conversations (additional service lines, ARKA-ED, the payer-side product) begin. By the *renewal*, the dashboard already shows cumulative dollars recovered, so the renewal is not a competitive evaluation—it is an accounting exercise. Each inflection deepens switching costs: the workflow is embedded, the residents trained on ARKA-ED, and the documentation that gets the hospital paid now flows through ARKA.

7.6 The Clinical-Champion Program

Adoption is not left to chance. Each deployment includes a structured six-month clinical-champion program: a respected local physician is identified and resourced to lead change management, monthly ROI reviews are held with the customer's CFO, and override patterns are reviewed with the champion to tune site-specific behavior. This is the human complement to the bandit's statistical tuning—the engine learns the site's baseline while the champion builds the cultural permission to trust it. The program directly attacks the single largest historical barrier to CDS retention: organizational change management, which most technology vendors discover they are doing for the first time only after the contract is signed.

Table 7.2. The six-month clinical-champion cadence.

Month	Focus
1	Identify champion; baseline metrics; configure site rules; kickoff
2--3	Shadow + soft launch; weekly override review; tune thresholds
4	First ROI review with CFO; publish internal win stories
5--6	Default-on with override; quarterly business review; expansion planning

7.7 Retention Metrics and Long-Term Outlook

Table 7.3. Retention and expansion metrics.

Metric	Target	Driver
Net revenue retention	115% → 124%	Expansion + multi-product adoption
Logo (contract) churn	<5% annually	Embedded workflow + demonstrated ROI
Physician engagement	>60% by Month 6	Zero workflow change + earned interruptions
RBM time-per-case reduction	~60%	Auto-approval + cited reasons
Health-system savings (Year 5)	~\$12,800/bed/yr	Compounding utilization improvement

The retention advantage. Once Arka achieves majority physician engagement, the system becomes self-reinforcing: it is embedded in the workflow, it is the source of the documentation that gets the hospital paid, it is the engine the residents trained on, and it is the dashboard the CFO checks for recovered dollars. Displacing it would mean re-introducing the friction Arka removed—which is why retention, not acquisition, is the deepest moat.

8 CHAPTER 8

Implementation Action Plan with Timeline

Arka's implementation plan sequences three operating phases against a parallel fundraising and engineering roadmap. The guiding principle is the one that defines the whole company: solve the workflow and the money first, build the technology last, and let each phase fund and de-risk the next.



Figure 8.1. The 36-month implementation and fundraising timeline.

Table 8.1. Phase-by-phase activities and success metrics.

Phase	Key activities	Success metric
Phase 1: Validation (M1--6)	Deploy ARKA-CLIN at 3–5 early adopters; stand up ROI dashboard; capture overrides; begin validation study	3–5 deployments; 1 publication submitted; documented ROI floor
Phase 2: Growth (M6--18)	Scale sales; sign 10–15 systems; launch ARKA-ED in 5 academic centers; standardize the 90-day pilot	15 deployments; \$2M ARR; Series A closed
Phase 3: Expansion (M18--36)	ARKA-INS payer pilots; scale ARKA-ED nationally; OPTIMA O-3→O-5	50+ deployments; 2–3 payers live; Series B readiness

8.1 Phase 1 --- Validation (Months 1--6)

The objective of Phase 1 is to validate the AIIE engine in a live clinical environment, demonstrate quantified ROI, and establish the first reference customers. Arka deploys ARKA-CLIN at 3–5 early-adopter health systems (150–300 beds and teaching hospitals), generates case studies, and publishes initial data. Activities include integrating via CDS Hooks and SMART-on-FHIR at each site, standing up the ROI dashboard, capturing override reasons, and beginning a peer-reviewed validation study. **Success metrics:** 3–5 deployments, one peer-reviewed publication submitted, and a documented ROI floor at each site.

8.2 Phase 2 --- Growth (Months 6--18)

Phase 2 scales the sales team and signs 10–15 health systems (500+ beds) and imaging centers in major metros, while launching ARKA-ED in five academic centers. The 90-day self-funding pilot becomes the standard sales motion. **Success metrics:** 15 deployments, \$2M ARR, ARKA-ED live in five academic centers, and Series A closed.

8.3 Phase 3 --- Expansion (Months 18--36)

Phase 3 expands into the payer market (ARKA-INS pilots with 2–3 payers, riding CMS-0057-F), scales ARKA-ED nationally, and targets 30% market share in core segments. **Success metrics:** 50+ hospital deployments, ARKA-INS in production with 2–3 payers, 30+ academic centers, and Series B readiness.

8.4 Fundraising Roadmap

Arka pursues a capital-efficient path enabled by its Non-Device posture: because the product ships as software rather than a cleared device, each round buys go-to-market and engineering rather than regulatory runway.

Table 8.2. Fundraising roadmap.

Round	Timing	Use of proceeds
Pre-seed / Seed	Months 0–6	MVP hardening; first 3–5 deployments; clinical advisory board
Series A (\$25M)	Months 12–18	GA of ARKA-CLIN; ARKA-ED build; AIIE-OPTIMA O-0→O-3; regional sales
Series B	Months 30–36	National scale; ARKA-INS payer rollout; OPTIMA O-4→O-5; international exploration

Pilot data showing quantified ROI and physician adoption is the centerpiece of the Series A narrative for healthcare-focused funds (a16z Bio+Health, General Catalyst, Flare Capital, Bessemer). At Series A, Arka will have proven clinical ROI across multiple sites, a live compliance package for CMS-0057-F, and a defensible Non-Device regulatory posture.

8.5 The AIIE-OPTIMA Engineering Roadmap

The engine roadmap (O-0 → O-5, detailed in Section 5.4) runs in parallel with the operating phases and is gated so that each milestone is proven before the next begins. Every phase remains Non-Device; the deterministic v1 core is the always-on fallback; and the rollout proceeds shadow → mirror → opt-in A/B → default-with-override. O-0 through O-3 are funded by the Series A; O-4 and O-5 (shadow-mode validation and the pre-registered efficacy readout) anchor the Series B story.

8.6 Milestones and Success Metrics

8.7 Team and Execution

Arka's execution risk is mitigated by a team built deliberately at the intersection of the three disciplines this problem requires: clinical informatics, machine learning, and healthcare revenue-cycle economics. The founding thesis—solve the workflow and the money first, build the technology last—is itself a hiring filter: Arka recruits people who

Table 8.3. Consolidated milestones by phase.

Milestone	Definition	Target month
First live deployment	ARKA-CLIN scoring real orders in production	M3
First publication submitted	Peer-reviewed appropriateness-lift study	M6
\$2M ARR	Across CLIN + ED	M18
ARKA-INS first payer pilot	Live PAS scoring on real authorizations	M22
OPTIMA O-3 complete	Federated network + DR-OPE validated	M24
50+ hospital deployments	Across all tiers	M36
Pre-registered OPTIMA readout	≥15% dose / ≥20% cost at non-inferior yield	M34

have shipped inside Epic/Cerner integrations, who have sat on the payer side of a prior-authorization desk, and who have built explainable ML in regulated settings. A clinical advisory board of practicing radiologists, emergency physicians, and primary-care clinicians governs the knowledge graph and signs off on every in-scope module. Key-person risk is managed through documented architecture, a deep engineering bench, and institutionalized review discipline (the boundary lint and clinical sign-off log are part of the codebase, not a single person's memory).

8.8 Quarter-by-Quarter Operating Plan

Table 8.4. Eight-quarter operating plan (post-Series A).

Quarter	Primary objectives
Q1	Close Series A; harden ARKA-CLIN GA; first 3 paid deployments live; OPTIMA O-0 scaffold
Q2	5–7 deployments; ROI dashboards live; submit first validation manuscript; OPTIMA O-1 (VOI + HGNN)
Q3	10 deployments; ARKA-ED pilots at 2 academic centers; first payer discovery calls; OPTIMA O-2
Q4	15 deployments; \$2M ARR; ARKA-ED in 5 centers; sign first payer pilot LOI; OPTIMA O-3 (federated + DR-OPE)
Q5	First ARKA-INS payer pilot live; expand CLIN into 2 new regions; second publication
Q6	25+ deployments; payer pilot expand-or-no-go → rollout; OPTIMA O-4 shadow at 3 sites
Q7	National ARKA-ED launch; 2nd–3rd payer pilots; begin Series B conversations
Q8	40+ deployments; pre-registered OPTIMA readout; OPTIMA O-5 default-with-override; Series B close

8.9 Organization and Hiring Plan

The Series A funds a focused build: a clinical-informatics and ML team for the engine, a standards-and-integration team for CDS Hooks / FHIR / Da Vinci, a regional sales and customer-success organization, and a small regulatory and clinical-validation function. Because Arka ships software on existing rails rather than clearing a device, headcount concentrates on go-to-market and engineering rather than regulatory affairs—the capital-efficiency advantage of the Non-Device posture, expressed as an org chart.

Table 8.5. Indicative functional headcount build by phase.

Function	Ph. 1	Ph. 2	Ph. 3	Mandate
Engineering & ML	10	22	45	AIIE-OPTIMA, CDS Hooks/FHIR, platform
Clinical & regulatory	4	8	14	Knowledge graph, validation, advisory board
Sales & customer success	5	16	32	Regional + payer GTM, 90-day pilots, renewals
Implementation & support	3	6	12	Integrations, deployments, champion program
G&A (finance, legal, ops)	3	7	12	Finance, contracts, security & compliance
Total FTE	25	55	110+	Reaching ~300 by Year 5

Cybersecurity Framework and Data Protection

Healthcare is the most-breached sector in the economy, and a single PHI incident can end a young company. Arka's security posture is therefore not a bolt-on compliance layer but an architectural choice made at the foundation: the system is designed so that the most sensitive data—raw patient records—never has to move at all.

9.1 The Healthcare Cybersecurity Landscape

Healthcare data is uniquely valuable to attackers and uniquely costly to lose: breach-remediation costs in healthcare run well above the cross-industry average, and the regulatory and reputational consequences are severe. The market response is a fast-growing healthcare-cybersecurity spend, compounding at roughly 17% per year—a tailwind for any vendor that can credibly claim privacy-by-architecture (Figure 9.1).

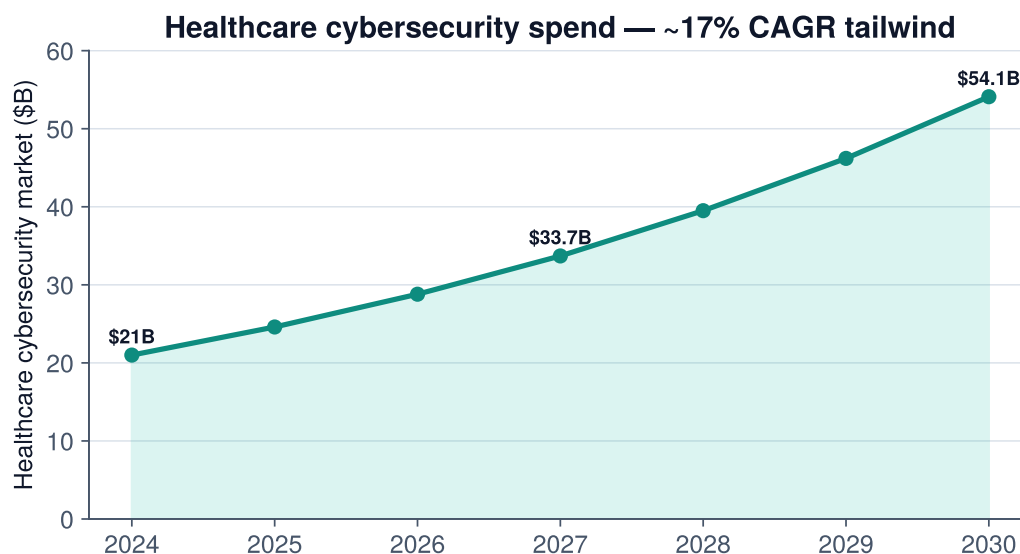


Figure 9.1. Healthcare cybersecurity spend—a ~17% CAGR tailwind.

9.2 ARKA's Security Architecture: Privacy by Design

Arka's defining security property is that it improves across institutions *without centralizing patient data*. The engine trains via federated learning with differential privacy: models travel to the data, only differentially private gradient aggregates return, and a per-institution ϵ -budget ledger bounds what any release can reveal. This eliminates the single largest category of healthcare breach—the centralized data lake—because there is no lake to breach.

Table 9.1. ARKA's defense-in-depth controls.

Control layer	Implementation
Data minimization	Consumes only the structured FHIR fields required to score an order; no images, no waveforms
Federated learning + DP	Raw PHI never leaves the institution; only DP gradient aggregates move ($\epsilon = 1.0$, $\delta = 10^{-6}$ /week)
Encryption	TLS 1.3 in transit; AES-256 at rest; per-tenant key isolation
Zero-trust access	Least-privilege, short-lived scoped tokens (SMART-on-FHIR); full authentication on every call
Log redaction	Identifiers reduced to salted hashes and age buckets before logging
Audit trail	Every recommendation, override, and data access is immutably logged for inspection

Table 9.2. Threat model and ARKA's structural mitigations.

Threat	Structural mitigation
Central data-lake breach	No lake exists—federated learning keeps records at the institution
PHI exfiltration in transit	TLS 1.3; only DP gradient aggregates ever traverse the network
Re-identification from aggregates	Differential-privacy ϵ -budget bounds information leakage
Credential compromise	Short-lived scoped SMART tokens; least-privilege; full audit
Ransomware	Immutable audit logs; tested IR plan; minimal local PHI footprint
Model inversion / extraction	Aggregate-only training; no patient-level outputs exposed

9.3 HIPAA Compliance and Breach Response

Arka operates as a HIPAA business associate, executing business-associate agreements with every covered-entity customer and adhering to the Privacy, Security, and Breach Notification Rules. The federated architecture dramatically narrows the breach surface, but Arka nonetheless maintains a formal incident-response program: detection and containment within defined SLAs, forensic analysis, individual notification for any breach affecting protected information, notification to HHS, and—for incidents affecting 500 or more individuals in a jurisdiction—the required media notification. Tabletop exercises are run quarterly, and the response plan is tested against realistic ransomware and exfiltration scenarios.

Table 9.3. Incident-response cadence and service levels.

Phase	Action & target
Detect	24/7 monitoring; anomaly alerts triaged within minutes
Contain	Isolate affected scope; revoke tokens; target containment <1 hour
Investigate	Forensic analysis of redacted logs; scope determination
Notify	Affected-individual, HHS, and (for 500+ in a jurisdiction) media notification within required windows
Remediate	Root-cause fix; control hardening; post-incident review

9.4 Regulatory and Certification Roadmap

Arka pursues the certifications that healthcare buyers' security reviews require, on a schedule aligned to the growth phases.

Table 9.4. Security and compliance certification roadmap.

Certification	Target	Why it matters
SOC 2 Type I	Phase 1	Minimum bar for hospital IT security review
SOC 2 Type II	Phase 2	Demonstrates sustained control operation over time
HITRUST CSF	Phase 2-3	The de facto trust standard for healthcare data partners
NIST 800-53 alignment	Ongoing	Federal-program and Tricare readiness
Penetration testing	Continuous	Independent third-party validation each release cycle

Cybersecurity key takeaway. Arka's strongest security control is architectural, not procedural: by never moving raw PHI and never analyzing images, Arka removes the two largest attack surfaces in medical-imaging software before a single firewall rule is written. Privacy-by-architecture is simultaneously a compliance posture, a sales accelerant (it clears security review faster), and a moat (the federated outcome dataset cannot be replicated without the same privacy primitives).

Roadblocks, Risk Mitigation, and Adaptation

Every venture of consequence faces structural risks. Arka's advantage is that most of the risks that kill health-tech startups were designed out of the company at the architectural level—and the ones that remain are explicitly registered, owned, and mitigated. This chapter catalogs the anticipated roadblocks, the formal risk register, and the technology-adaptation strategy that keeps Arka ahead of the curve.

10.1 Anticipated Roadblocks and Mitigation Strategies

A few of these warrant emphasis. **EHR-vendor competition** is the most-cited investor concern; the answer is that Arka sits *on top* of Epic and Cerner as a standard CDS-Hooks app and occupies the bilateral provider-plus-payer position that an EHR's generic best-practice alerts structurally cannot. **Budget constraints** are answered not by discounting but by reframing: Arka is recovered revenue with a software invoice attached, and the ROI dashboard makes that case monthly to the CFO who signs the renewal.

FDA reclassification is the existential tail risk for any clinical-AI company. Arka manages it not with optimism but with engineering: the hard NO-list is enforced by a continuous-integration boundary lint that fails any build importing an out-of-scope capability, a Q-Submission pre-submission is planned to obtain FDA's written concurrence, and the one feature that would cross the line—Epoch 2 autonomous triage—is quarantined out of the roadmap entirely. **Alert fatigue** is the most common reason CDS is disabled within a quarter; Arka answers it with empty cards when no guideline fires, a per-site bandit that converges from ~15% to ~2% exploration, and a Value-of-Information floor that suppresses low-yield prompts. **Long payer sales cycles** are the runway killer; Arka de-risks them with a regional and RBM wedge that closes fast, the CMS-0057-F forcing function that is closing the deal regardless, and a 90-day pilot engineered to be cheaper than the buyer's current quarterly PA audit—so the pilot funds itself.

10.2 The Seven-Risk Register

Beyond the operational roadblocks, Arka maintains a formal seven-risk register scored on likelihood and impact, each with a named mitigation and a residual rating (Figure 10.1).

10.3 Technology Adaptation and Future-Proofing

Arka is architected to absorb change rather than be disrupted by it. Three design choices provide durable adaptability. First, **standards-native integration** means that as EHR vendors evolve, Arka rides the same CDS Hooks / FHIR / Da Vinci rails the whole industry is converging on, rather than maintaining brittle proprietary connectors. Second, the **engine is decoupled from the interface**: AIIE-OPTIMA can be upgraded behind a stable `scoreOrder()` contract without any clinician seeing a new screen, so the science can advance continuously without re-triggering adoption risk. Third, **imaging is only the wedge**: the same sequential-decision engine that reasons over imaging pathways

Table 10.1. The ten anticipated roadblocks and ARKA's mitigations.

Roadblock	Mitigation
1. FDA regulatory reclassification	Hard NO-list + CI boundary lint; planned Q-Submission pre-submission; Epoch 2 quarantined; deterministic core preserves Criterion 4
2. Physician adoption / alert fatigue	Non-blocking, silent-unless-cited design; per-site bandit; 6-month clinical-champion change-management program
3. EHR-vendor in-house competition (Epic, Oracle Health)	Layer <i>on top</i> of incumbents via standards; bilateral position none of them hold; guideline-cited explainability they lack
4. RBM market consolidation	Position as a margin-raising layer for RBMs, not a rival; per-case economics invert once 35–40% auto-approve
5. Data breach / HIPAA enforcement	Federated DP; no raw PHI moves; redacted logs; ϵ -budget ledger; SOC 2 / HITRUST roadmap
6. Hospital budget constraints	“ARKA pays for itself” messaging; denial-recovery floor; monthly ROI review with the customer CFO
7. Crowded imaging-CDS market	Differentiate on bilateral engine, denial-recovery ROI, and cross-modality optimization; orthogonal to clinical-only competitors
8. Founder dependency / key-person risk	Deep bench; documented architecture; clinical advisory board; institutionalized engineering discipline
9. Malpractice / CDS-related litigation	Non-blocking recommendation-only posture; full audit trail; tech E&O coverage; specialized counsel retained
10. State-level regulatory patchwork	Configurable per-jurisdiction rules; cited reasons satisfy gold-card statutes; telehealth-rule monitoring

will, in 3–5 years, reason over specialty-drug step therapy, elective-surgery appropriateness, and any expensive medical decision against the evidence—the \$10B+ appropriateness layer of American medicine.

10.4 AI/ML Integration Roadmap: Optimization First, Intelligence Later

Arka's machine-learning strategy is deliberately staged to manage both technical and regulatory risk: *optimization first, intelligence later*. The deterministic, guideline-anchored core ships first and remains the always-on fallback; ML is layered in only to *refine* recommendations a guideline already supports, never to originate them. AIIE-OPTIMA's research-grade modules (the POMDP, MCTS, federated learning, and DR-OPE) are introduced behind a feature flag with statistical gates and automatic rollback, so that capability advances without ever compromising the explainability and Non-Device posture that adoption depends on.



Figure 10.1. The seven-risk register—all mitigated to acceptable residual.

Table 10.2. The seven-risk register in detail.

Risk	Likelihood	Mitigation
FDA reclassifies a feature as a device	Low–Med	NO-list + boundary lint; Q-Sub pre-submission; Epoch 2 excluded
Thin clinical validation (synthetic data)	Med	Rules-first/citation-first basis; retrospective validation + DR-OPE; clinical sign-off log
EHR integration / Epic dependency	Med	Standards-native (CDS Hooks/S-MART/Da Vinci); multi-EHR by design
Incumbent retaliation (Epic CDS, eviCore)	Med	Layer on top, not a rival; bilateral position none hold
Long payer sales cycle burns runway	Med–High	Regional/RBM wedge; CMS-0057-F forcing function; 90-day self-funding pilot
Data / privacy breach	Low	Federated DP; no raw PHI moves; redacted logs; ϵ -ledger
Model drift / alert fatigue	Med	Empty-cards-when-silent; per-site bandit; VOI floor; weekly DR-OPE auto-rollback

Table 10.3. The phased AI/ML capability roadmap—optimization first, intelligence later.

Stage	Capability	Guardrail
Now (v1)	Deterministic rules + GRADE weights + gradient-boosted refinement + SHAP	Always-on fallback; fully explainable
O-1--O-2	Bayesian VOI belief + MCTS pathway reasoning + HGNN encoder	Structured facts only; ≥ 2 options; click to act
O-3	Federated learning + DR-OPE continuous validation	DP ϵ -ledger; weekly auto-rollback
O-4--O-5	Shadow $\rightarrow$ A/B $\rightarrow$ default-with-override at scale	Pre-registered efficacy gate; still Non-Device
Excluded	Autonomous triage (Epoch 2)	Would require 510(k); not pursued

The discipline that bounds the roadmap. The single line Arka will not cross is autonomous action. Every recommendation, in every phase of the AI roadmap, requires an affirmative clinician click and exposes its evidentiary basis. The most advanced version of the engine is therefore not the one that decides for the clinician—it is the one that gives the clinician the most precisely reasoned, most transparently justified recommendation in under 800 milliseconds. *Optimization first, intelligence later, autonomy never.*

Appendices

11.1 Appendix A --- Assumptions and Sources for Every Number

Every dollar figure in this plan is a modeled, sourced range rather than a point guarantee. The provider model uses the conservative column throughout; the aggressive case is approximately $1.5\times$. These are decision-support economics, not a guarantee of clinical outcomes.

11.2 Appendix B --- The ROI Model, Line by Line

Provider side (regional group, 120,000 advanced studies/yr). Derivation of denial-recovery Lever A: $120,000 \times 0.25$ (PA-required) = 30,000 PAs $\rightarrow \times 0.25$ (denied) = 7,500 denials $\rightarrow \times 0.86$ (avoidable) = 6,450 $\rightarrow \times 0.60$ (ARKA-captured) = 3,870 recovered claims $\rightarrow \times \$900 = \3.48M .

Payer side (per 1,000 lives/yr, conservative). PA admin saved \$31; appeal-volume reduction \$15; appeal-overtake loss reduction \$28; inappropriate-imaging avoidance \$1,512; site-of-service leakage capture \$14,000; cascade-cost avoidance \$76; FWA reduction \$5. Gross $\approx \$15,667$; less \$0.40 PMPM fee (\$4,800); net $\approx \$10,867$ ($\approx 2.3\times$ year-1 ROI). Scales to $\sim \$543\text{M}$ net year-1 for a 50M-life national carrier (conservative), $\sim \$815\text{M}$ aggressive.

11.3 Appendix C --- OPTIMA Components versus the Four Statutory Criteria

Every shipped component satisfies all four criteria; only Epoch 2 (autonomous triage) fails Criteria 3 and 4 and is therefore excluded as a Class II SaMD path.

11.4 Appendix D --- Glossary of Terms

AIIE	ARKA Imaging Intelligence Engine—the shared reasoning core beneath all three products.
AIIE-OPTIMA	The next-generation sequential-decision engine (POMDP, MCTS, VOI, federated, DR-OPE).
CDS Hooks	HL7 standard that fires decision support at clinical decision points inside the EHR.
CMS-0057-F	The CMS Interoperability and Prior Authorization Final Rule; mandates a FHIR PA API by Jan 2027.
Da Vinci CRD/DTR/PAS	HL7 implementation guides for coverage requirements, documentation, and prior auth.
DR-OPE	Doubly-Robust Off-Policy Evaluation—continuous statistical proof of value with auto-rollback.
Federated learning	Training where data lives; only privacy-preserving aggregates move, never raw PHI.
GRADE	The international evidence-grading framework (Levels A–D) used to weight the knowledge graph.

Table 11.1. Assumptions and sources.

Figure	Value	Source
Imaging prior-auth denial rate	20–40%	Provider-reported; AHA/Premier 2024
Share of denials avoidable	~86%	Change Healthcare Denials Index
Denied claims never reworked	~50%	MGMA / rev-cycle reporting
Cost to rework a denied claim	\$25–\$118	MGMA
Appeal overturn rate (MA, 2024)	80.7%	KFF (Jan 2026)
MA PA determinations / denial / appeal (2024)	53M / 6.4% / 11.5%	KFF (Jan 2026)
US PA administrative spend	~\$13B/yr	CAQH 2024 Index
Cost per PA (manual / partial / electronic)	\$11 / \$6 / \$1.88	CAQH 2024 Index
Inappropriate imaging rate	30–35%	Choosing Wisely, ACR, Health Affairs
Redundant imaging within 90 days	15–20%	Peer-reviewed; AJR
Avg net reimbursement, advanced study	~\$900 (\$450–\$2,155)	Medicare allowed + commercial mix; HFMA
Incidental-finding cascade per finding	\$460–\$2,155	PubMed 30300007; JAMA Network Open
Low-value imaging / cascade (national)	\$12–30B / \$3–6B per yr	ACR; Yale GHR
Provider–payer disagreement	\$40B/yr	Internal synthesis of published data
Site-of-service leakage	~\$14M/yr per 1M lives	HFMA; NIHCR
STAR point value (500K-life plan)	\$80–120M/yr	KFF; payer analyses
Gold-Card / auto-approve eligibility	35–45%	UnitedHealthcare 2024 program data
AIIE capture rate of at-risk orders	60–72%	ARKA-CLIN deployment data (internal)
Physician PA burden	~39 PAs & ~13 hrs/week	AMA Prior Authorization Survey
CMS-0057-F est. 10-yr industry savings	\$15B	CMS 2024 Final Rule
PAMA AUC program	Rescinded (CY2024)	CMS; Reed Smith analysis
ARKA fee	\$0.30–\$0.50 PMPM / \$1.50–\$3.00 per PA	ARKA pricing

Table 11.2. Each OPTIMA component, verified against the four §520(o)(1)(E) criteria.

Component	C1 no img/sig	C2 evidence	C3 recommend	C4 review
Bayesian VOI kernel (D19)	yes	yes	yes	yes
Trajectory simulator (D20 engine)	yes	yes	yes (fenced)	yes
MCTS pathway planner (D20)	yes	yes	≥2 options	yes
Contextual bandit	yes	yes	surfaces only	yes
HGNN encoder	yes	yes	n/a	yes
Federated learning + DP (D21)	yes	yes	n/a	yes
DR-OPE value proof (D22)	yes	yes	n/a	yes
Auditable objective weights (D23)	yes	yes	n/a	yes
Warm-start bootstrap (D24)	yes	yes	yes	yes
Epoch 2 autonomous triage	yes	yes	NO	NO

MCTS Monte-Carlo Tree Search—ranks imaging *pathways*, not just single studies.

Non-Device CDS Software excluded from device regulation under FD&C §520(o)(1)(E).

PAMA AUC The (now-rescinded) Appropriate Use Criteria mandate under the Protecting Access to Medicare Act.

SHAP SHapley Additive exPlanations—the per-factor attribution behind every AIIE score.

VOI Value of Information—the expected diagnostic information a study yields, in bits.

11.5 Appendix E --- Brand: Gungnir and “remARKAbly precise”

The brand thesis is **Gungnir**—Odin’s spear that never misses. One platform, one standard of care, precision you can trust. The tagline *remARKAbly precise* carries four meanings at once: precise at the *math* (a POMDP that quantifies the value of information), the *architecture* (explainable, federated, continuously validated), the *regulatory posture* (Non-Device by design, Epoch 2 left on the shelf), and the *execution* (fast and capital-efficient). The revenue-first message leads with money—“You did the scan. Now get paid for it.”—and uses the precision thesis as the proof beneath the dollar claim. Precision is why the claim clears; getting paid is why the CFO signs.

Closing. Arka is the appropriateness layer of American medicine, beginning with imaging. It gets hospitals paid, passes every regulation by design, never touches the doctor’s workflow, and proves its own value continuously and statistically. The bilateral engine is the wedge; the sequential-decision engine is the moat; staying Non-Device is the strategy; getting the hospital paid is the pitch. *remARKAbly precise*.

11.6 Appendix F --- Frequently Asked Investor Questions

Isn’t this just another clinical-decision-support tool, and didn’t those all fail?

The CDS graveyard is full of tools that interrupted the physician and billed the hospital for the privilege. Arka inverts both: it leads with the buyer’s money (denial recovery) and never touches the workflow (non-blocking, sub-800ms,

silent unless a guideline fires). It is also bilateral—the same engine serves the payer—which no failed single-sided CDS tool ever was.

Will Epic or Oracle Health just build this?

EHR best-practice alerts are generic, not appropriateness-grade, and payer-blind. Arka occupies the bilateral provider-plus-payer position from a single math, cites published guidelines, and exposes SHAP explanations—and it rides on the EHR's own CDS Hooks rails as a layer *on top*, not a rival. An EHR could build a worse version of half of it.

What happens to the thesis now that PAMA AUC was repealed?

The repeal removed the regulatory *stick* that would have entrenched incumbents and forced consultation of decision support. Adoption now rides on value and culture—which is precisely the revenue-first, education-led space Arka occupies. Meanwhile CMS-0057-F adds a far stronger forcing function on the payer side. The net of the four-event earthquake is favorable.

Is the ML defensible, or is it a black box?

The primary basis of every recommendation is a published guideline cited above any ARKA score, and the deterministic core can render the full decision with no ML at all. ML only refines; it never originates. That is what keeps Arka inside Non-Device CDS and what makes the recommendations defensible in an appeal or an audit.

How do you avoid the 510(k) trap that slows imaging-AI companies?

By never analyzing a pixel, never predicting a time-critical event, and never acting autonomously. Arka consumes only structured FHIR and enforces this with a CI boundary lint. OPTIMA Epoch 2 (autonomous triage) is deliberately excluded precisely because it would force the 510(k) path.

What's the realistic downside?

The conservative denial-recovery floor (\$3.7–4.0M net Year-1 per regional group) requires no utilization-reduction assumption at all—only that the hospital collect revenue it already earned. Even the worst-case ROI sensitivity corner clears 450%, and the worst-case three-phase revenue scenario still delivers \$64.9M in Year 5. The downside is bounded by design.

Why is now the right time?

CMS-0057-F's FHIR PA API is mandatory in January 2027; the FDA broadened the Non-Device boundary in January 2026; PAMA AUC was repealed; and AHIP plus state gold-card laws are raising the compliance bar. The technology (CDS Hooks, FHIR R4, Da Vinci) is mature, and Arka ships the mandated endpoint today. The window is open and time-boxed.

What does the Series A actually buy?

General availability of ARKA-CLIN, the ARKA-ED build, AIIE-OPTIMA phases O-0 through O-3, and a regional sales and customer-success organization—enough to reach \$2M ARR, multiple peer-reviewed publications, and Series B readiness. Because the product ships as software, capital funds growth rather than regulatory runway.

ARKA is an FDA Non-Device Clinical Decision Support tool under §520(o)(1)(E) of the FD&C Act (21st Century Cures Act §3060), as clarified in FDA's CDS Software Final Guidance (January 6, 2026). ARKA reasons over structured clinical data and published evidence; it does not analyze medical images or signals, does not predict time-critical clinical events, and does not place orders autonomously. The ordering clinician retains full responsibility for the final decision. Dollar figures are modeled, sourced ranges—decision-support economics, not a guarantee of outcomes.

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